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# A TabNet-SHAP framework with stability-weighted multi-method feature selection for muscle injury prediction in professional football

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**Muscle injuries account for approximately 31% of all time-loss injuries in professional football, yet existing prediction models are constrained by small sample sizes, dependence on proprietary physiological data, limited interpretability, and inadequate temporal validation designs. This study develops an interpretable, strictly prospective prediction framework trained on 438,219 player-season records sourced from Transfermarkt, spanning 15 seasons (2010/11–2024/25) across 32 leagues. To ensure temporal integrity, all predictive features are constructed exclusively from prior-season data, and models are evaluated under a chronological hold-out design in which earlier seasons serve as training data and the two most recent seasons constitute the test set. We propose a multi-method stability-weighted feature selection (MSFS) strategy that cascades filter-based redundancy removal, four-method ensemble voting, and cross-validation stability verification, reducing 57 candidate features to a robust subset. Under cost-sensitive learning designed for severe class imbalance (6.2% positive rate), TabNet attains the highest discrimination among five candidate classifiers with an area under the receiver operating characteristic curve (AUC) of 0.812 (95% CI: 0.804–0.820), outperforming XGBoost (0.806), LightGBM (0.803), long short-term memory network (LSTM) (0.789), and logistic regression (0.761). Ablation experiments confirm that stability-verified feature selection and metaheuristic hyperparameter optimisation yield complementary performance gains, and direct comparison with Bayesian optimisation demonstrates that the detective behaviour algorithm provides competitive tuning across architectures. A seven-layer Shapley additive explanations (SHAP) interpretability analysis reveals that under the prospective design, injury history emerges as the most important feature category (33%), surpassing market-and-career features (29%), confirming that the temporal correction elevates genuine recurrence risk over reporting-related proxies. A two-stage risk architecture is identified in which market-and-career features establish a baseline risk stratum, whereas historical injury burden acts as a multiplicative amplifier among predisposed individuals. Individual-level decompositions further indicate that substitution-based rotation patterns are associated with attenuated age-related risk elevation. Sensitivity analyses excluding market value features quantify the extent to which predictive performance reflects genuine injury risk versus differential reporting completeness across leagues. These findings demonstrate that publicly accessible transfer-market data, combined with attention-based deep tabular learning and rigorous temporal validation, can support meaningful and interpretable injury risk stratification without proprietary monitoring systems.**

**Keywords** Muscle injury prediction, Professional football, Deep tabular learning, Explainable artificial intelligence, Feature selection stability, Shapley additive explanations, Class imbalance, TabNet, Temporal validation

## Background

Muscle injuries are the most prevalent injury category in professional football, accounting for approximately 31% of all time-loss injuries and representing the leading cause of player unavailability across elite European

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leagues[1]. Epidemiological evidence has consistently shown that fixture congestion, age-related degeneration, and prior injury history significantly elevate muscle injury risk, with match-related incidence rates far exceeding those observed during training[2]. Given the substantial medical, financial, and competitive consequences of muscle injuries, developing accurate and actionable prediction tools remains a strategic priority for the football industry[3].

## Machine learning for sports injury prediction

Machine learning (ML) has emerged as a promising paradigm for sports injury risk prediction, offering the capacity to model complex, nonlinear relationships among risk factors that traditional statistical approaches may fail to capture. Several systematic reviews have consolidated the evidence base. Claudino et al.[4] surveyed artificial intelligence applications for injury risk assessment across team sports, identifying a growing but methodologically heterogeneous body of work. Van Eetvelde et al.[5] found that tree-based ensemble methods—random forests and gradient boosting machines—consistently ranked among the best-performing algorithms, with area under the receiver operating characteristic curve (AUC) values typically ranging from 0.70 to 0.85. More recently, Leckey et al.[6] confirmed that while ML models demonstrate statistical promise, predictive performance varies substantially depending on data quality and study design, and Yuan et al.[7] highlighted the persistent gap between reported model accuracy and real-world utility.

Instructive results have also been obtained in other sports. Pareek et al.[8] applied XGBoost to publicly available data from 2,103 National Basketball Association players spanning 20 seasons to predict lower-extremity muscle strains, achieving an AUC of 0.84 with prior injury history and age identified as dominant predictors. Jauhiainen et al.[9] reported more modest discrimination (AUC 0.63–0.65) using regularised logistic regression in a youth athlete cohort, while Cohan et al.[10] demonstrated the potential of deep sequential models for capturing temporal workload patterns in basketball injury forecasting. These cross-sport findings suggest that ML can extract meaningful risk signals from structured athletic data, while also revealing persistent challenges in sample size, external validation, and interpretability[6]. While these basketball-based studies demonstrate the feasibility of ML-driven injury prediction using publicly available data, football presents distinct challenges: larger and more heterogeneous squad compositions, more complex competition calendars spanning domestic leagues and continental tournaments simultaneously, and substantially greater variation in injury reporting practices across leagues and tiers[1]. These football-specific challenges motivate the development of a dedicated framework capable of handling the scale and heterogeneity of multi-league public data.

## Machine learning for football injury prediction

Within football specifically, ML-based injury prediction has attracted sustained research attention, producing a diverse but fragmented literature. From a data perspective, the majority of studies rely on proprietary global positioning system (GPS) tracking and workload monitoring data accessible only to individual clubs or research consortia. Rossi et al.[11] pioneered the application of recurrent neural networks to GPS and training-load data from a single professional club, achieving moderate predictive accuracy. Majumdar et al.[12] reviewed this literature comprehensively, noting that external load indicators—high-speed running distance, acceleration counts, and sprint frequency—constitute the most commonly employed predictive features. Freitas et al.[13] compared multiple algorithms for non-contact injury prediction in a professional squad, reporting accuracy around 0.74 with sensitivities of 0.71–0.77, though precision remained low (0.10–0.13), illustrating the challenge of excessive false positives under class imbalance.

Methodologically, the field has progressively incorporated more sophisticated architectures. Rommers et al.[14] applied tree-based models to pre-season physical screening data in elite youth players, achieving accuracies of 0.78–0.85. Calderón-Díaz et al.[15] targeted muscle injury prediction using biomechanical features combined with explainable ML techniques. Deep learning approaches have also been explored: Ye et al.[16] proposed a time-series image encoding method with deep autoencoders, achieving AUC 0.89 and sensitivity 0.82, while Martins et al.[17] constructed a four-year longitudinal prediction framework for muscle injury risk.

Despite this methodological diversity, studies utilising publicly available data remain rare and severely limited in scale. Diniz et al.[18] analysed match statistics for 209 Brazilian league players, with random forests achieving AUC 0.74. Kampakis[19] explored decision trees on a small professional cohort. More recently, Ma et al.[20] applied support vector machines to a dataset of 800 university-level players and reported an AUC of 0.99—a value the authors acknowledged as an artefact of artificial 1:1 class balancing rather than genuine predictive capability. These studies collectively demonstrate that while public data offer advantages in accessibility and reproducibility, no existing work has leveraged large-scale publicly available data to construct a robust and generalisable prediction framework.

## Interpretability, feature selection, and deep tabular learning

The development of explainable and methodologically rigorous ML frameworks spans advances in feature selection theory, model interpretability, and deep tabular learning. In feature selection, Saeys et al.[21] introduced the ensemble paradigm, demonstrating that aggregating results across multiple selection methods substantially improves stability without sacrificing predictive performance. Nogueira et al.[22] subsequently formalised the measurement of selection stability, establishing theoretical frameworks for quantifying reproducibility across data perturbations. Khaire and Dhanalakshmi[23] reviewed stability metrics comprehensively, while Pes[24] provided cross-domain empirical evidence that ensemble strategies consistently outperform individual methods. Despite these advances, feature selection in sports injury prediction has relied almost exclusively on single methods—such as random forest importance or Lasso alone—without cross-method validation or stability assessment. Recent advances in feature selection for high-dimensional binary classification have proposed several methodological innovations relevant to this challenge. Naeem et al.[45] introduced a greedy heuristic fuzzy clustering approach

(GHFClust) that combines interquartile-range-based feature screening with fuzzy-c-means clustering to handle dimensionality reduction in microarray data, achieving higher accuracy than single-method baselines. Ali et al.[46] proposed a weighted Fisher score (WFISH) that assigns differential weights to features based on class-discriminative expression patterns, demonstrating that weighting schemes can substantially improve feature ranking beyond unweighted univariate filters. Particularly relevant to the class imbalance challenge in injury prediction, Gul et al.[47] developed a margin-weighted robust discriminant score (MW-RDS) that integrates minority-class amplification factors into the feature ranking process, addressing the tendency of conventional selection methods to favour majority-class patterns. These developments underscore the need for multi-criteria, stability-aware feature selection strategies in imbalanced high-dimensional settings—a need that the present study addresses through the MSFS pipeline.

In model interpretability, Lundberg and Lee[25] proposed Shapley additive explanations (SHAP), a unified framework grounded in cooperative game theory that provides both local and global explanations. SHAP rapidly gained adoption in clinical prediction contexts; for instance, Nohara et al.[26] systematically demonstrated its value for real-world hospital data. Concurrently, Arik and Pfister[27] introduced TabNet, a sequential attention-based architecture for tabular data that performs instance-wise feature selection through learnable sparse masks, offering intrinsic interpretability alongside competitive accuracy. Oba et al.[28] applied TabNet to diabetes prediction, leveraging attention mask visualisation for transparent decision support. Benchmark studies have established that gradient-boosted trees generally remain competitive on medium-dimensional tabular tasks[29], though deep architectures may excel with large samples and complex feature interactions[30]. More broadly, attention-based architectures with quantitative regularisation strategies have demonstrated robust classification performance across diverse data modalities[49]. Nguyen and Byeon[31] exemplified this potential by combining TabNet with SHAP for cardiac arrest survival prediction, achieving AUC 0.993 with transparent feature attributions.

In sports science, SHAP-based interpretability has expanded rapidly in recent years. Moustakidis et al.[32] leveraged SHAP to identify key performance metrics in football team prediction, and Yan et al.[33] applied SHAP to professional basketball game analysis. Within football injury prediction specifically, however, SHAP applications remain limited. Calderón-Díaz et al.[15] and Jiacheng et al.[34] represent the current state of the art, yet both confine their analysis to single-layer global feature importance plots without exploring nonlinear dose-response relationships, feature interactions, or individual decision pathways. This limited utilisation of SHAP leaves a gap between model outputs and the multi-granularity, case-specific insights that practitioners require for informed decision-making.

## Research gaps

Four interrelated limitations characterise the current state of football injury prediction research. First, data accessibility and sample scale severely constrain reproducibility and generalisability: the predominant reliance on proprietary data restricts studies to single-club settings[11], while the few public-data studies involve sample sizes of 200–800 players[18]. Second, methodological approaches to class imbalance remain problematic, as many studies apply synthetic oversampling techniques that risk introducing distributional artefacts[35]; the AUC of 0.99 reported under artificial class balancing[20] illustrates how such practices can generate misleading performance benchmarks. Third, both feature selection and interpretability analysis lack sufficient rigour: feature selection relies on single methods without stability verification[22], while SHAP is typically confined to global feature importance bar plots[15], missing the multi-layered analysis needed for actionable insights. Fourth, temporal validation designs are largely absent from the existing literature. Most studies employ random train-test splits that allow future observations to inform past predictions, risking data leakage when same-season performance statistics are used to predict same-season injuries[6]. This concern is particularly acute for workload-related features—such as match density and playing time—which may be consequences rather than antecedents of injury within the same season. A strictly prospective design using only prior-season data for feature construction, combined with chronological hold-out validation, is essential to establish genuine predictive utility rather than retrospective correlation.

## Contributions

To address these gaps, this study proposes an interpretable TabNet-based prediction framework for professional football muscle injuries, trained on 438,219 player-seasons from Transfermarkt spanning 15 seasons (2010/11–2024/25) across 32 leagues. The framework integrates a strictly prospective prediction design, a multi-method stability-weighted feature selection (MSFS) strategy, cost-sensitive learning, and a seven-layer SHAP interpretability analysis. The principal contributions are:

- We propose an interpretable TabNet-based injury prediction framework that integrates cost-sensitive learning with attention-based deep tabular modelling on large-scale public data, underpinned by a strictly prospective prediction design in which all features are constructed exclusively from prior-season data and models are evaluated under chronological train-test partitioning. Five classifiers and multiple hyperparameter optimisation strategies are systematically compared to establish a scalable and reproducible alternative to proprietary monitoring approaches.
- We develop an MSFS strategy that combines filter-based redundancy removal, four-method ensemble voting, and cross-validation stability verification, producing a robust feature subset with demonstrated cross-method agreement and cross-fold consistency.
- We implement a seven-layer SHAP interpretability analysis that extends beyond conventional single-plot approaches to provide multi-granularity insights spanning global importance, nonlinear dependence, pair-

wise interactions, and individual decision pathways, enabling both population-level risk identification and case-specific interpretation.
- Through extensive experiments on the largest publicly available football injury dataset to date, including sensitivity analyses that quantify the contribution of reporting bias to predictive performance, we demonstrate that the proposed framework achieves robust discrimination and reveal a two-stage risk architecture in which market-and-career features establish baseline risk while historical injury burden acts as a multiplicative amplifier, providing actionable insights for squad management and injury prevention.

## Methods

### Overall framework

The methodological pipeline comprises five sequential stages designed to transform raw public transfer-market data into interpretable injury risk predictions (Fig. 1). *Data collection and preprocessing* involves acquiring player-season records from Transfermarkt, performing quality filtering, handling missing values, and addressing class imbalance through cost-sensitive weighting. *Temporal feature engineering* constructs 57 candidate features across six thematic categories, with all features derived exclusively from prior-season data to ensure strict temporal precedence between predictors and outcomes. *Multi-method stability-weighted feature selection* applies a three-stage pipeline—filter-based redundancy removal, four-method ensemble voting, and cross-validation stability verification—to reduce the feature space to a robust subset. *Model training and comparison* systematically evaluates five candidate classifiers under cost-sensitive learning, with hyperparameters optimised via both the detective behaviour algorithm (DBA) and Bayesian optimisation for fair comparison. *SHAP interpretability analysis* applies a seven-layer analytical framework to the best-performing model, providing multi-granularity insights into the predictive mechanisms driving muscle injury risk. In addition, supplementary sensitivity analyses quantify the influence of reporting bias through market value ablation and league-tier stratification.

The entire pipeline from data preprocessing through feature selection, model training, hyperparameter optimisation, and SHAP analysis is fully automated with no manual tuning at any stage. All hyperparameters are determined algorithmically: filter thresholds are fixed a priori, feature selection methods use cross-validated regularisation paths, and model hyperparameters are optimised via metaheuristic or Bayesian search. The complete pipeline is implemented in Python 3.10 and will be made publicly available upon acceptance.

### Data collection, preprocessing, and class imbalance handling

Data were collected from Transfermarkt (https://www.transfermarkt.com), a publicly accessible platform providing comprehensive football player career statistics, transfer histories, market valuations, and injury records. The raw dataset spans 19 consecutive seasons from 2006/07 to 2024/25, covering 32 professional leagues across 14 countries; the additional early seasons (2006/07–2009/10) serve exclusively as sources of prior-season and cumulative historical features for the analysis period beginning in 2010/11 and do not enter the modelling sample. The primary analysis encompasses 15 seasons from 2010/11 to 2024/25. Seasons prior to 2010/11 were excluded because Transfermarkt's injury reporting coverage was demonstrably sparse in the earlier period (e.g., only 68 muscle injuries recorded across all leagues in 2006/07), which would introduce systematic ascertainment bias rather than reflecting true injury incidence.

Each observation corresponds to a unique player-season, defined as a player's aggregated record within a single competitive season. For players who transferred mid-season, records were aggregated at the player-season level by summing appearance-related statistics across clubs within the same season, while market valuations were taken from the player's primary club (defined as the club where the player recorded the most appearances).

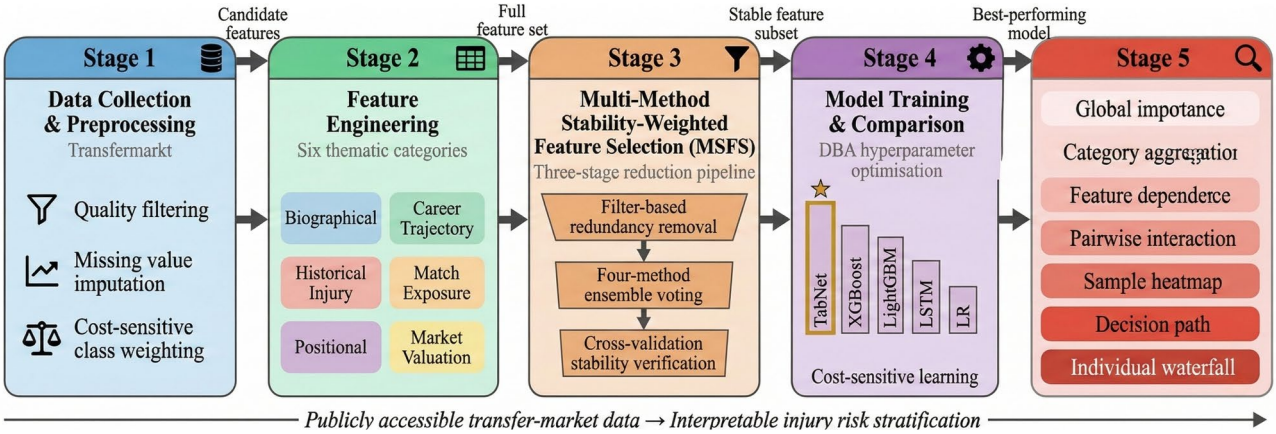



**Fig. 1**. Overview of the proposed methodological framework. The pipeline proceeds through five sequential stages: data collection and preprocessing from Transfermarkt, feature engineering across six thematic categories, multi-method stability-weighted feature selection (MSFS), model training and comparison with detective behaviour algorithm (DBA) hyperparameter optimisation, and seven-layer SHAP interpretability analysis.

| Threshold | Player-seasons | Positive rate (%) | AUC-ROC | AUC-PR |
|---|---|---|---|---|
| $\geq$ 1 appearance | 467,814 | 5.83 | 0.808 [0.800–0.816] | 0.268 |
| $\geq$ 3 appearances (adopted) | 438,219 | 6.20 | 0.812 [0.804–0.820] | 0.274 |
| $\geq$ 5 appearances | 412,637 | 6.51 | 0.814 [0.806–0.822] | 0.279 |
| $\geq$ 10 appearances | 358,491 | 7.08 | 0.810 [0.801–0.819] | 0.272 |

**Table 1**. Sensitivity analysis of the minimum appearance threshold on dataset size, class balance, and downstream model performance (TabNet + MSFS + DBA, temporal hold-out).

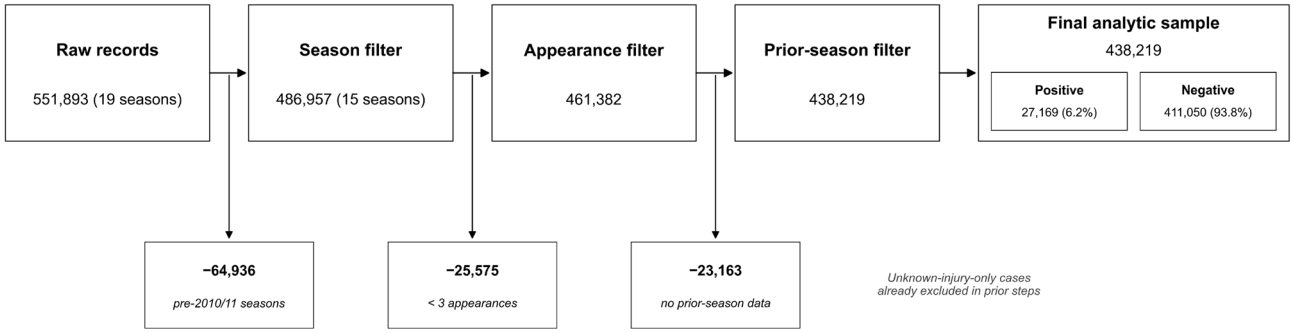



**Fig. 2**. Data quality filtering pipeline. Each box reports the remaining sample size after the corresponding exclusion criterion. Player-seasons with only "unknown injury" records are reclassified as negative rather than excluded, preserving the sample size. The final analytic sample comprises 438,219 player-seasons, of which 27,169 (6.2%) are positive (at least one muscle injury during season $t$) and 411,050 (93.8%) are negative.

Duplicate entries arising from players registered with multiple clubs in the same season were merged using the player identifier as the primary key.

The binary target variable $y \in \{0, 1\}$ indicates whether a player sustained at least one muscle injury during season $t$, identified through keyword-based classification of Transfermarkt's free-text injury descriptions. Injury records were classified as muscle injuries if the description contained any of the following terms: "muscle," "muscular," "hamstring," "adductor," "groin," "calf," "thigh," "strain" (excluding non-muscle strains such as "ankle strain"), "torn muscle fiber," or "pulled muscle." Player-seasons in which the only recorded injuries were classified as "unknown injury" ($n = 27{,}028$ across the full 19-season dataset) were excluded from positive-class assignment, as their ambiguous classification could not reliably be attributed to either the muscle injury or non-injury class. Within the final analytic sample, any such cases that survived the quality filters in Steps 1–4 below are treated as negative (no documented muscle injury).

*Data quality filtering*

Data preprocessing proceeded through five steps, with sample counts reported at each stage to ensure transparency:

1. Raw records: 551,893 player-seasons across 19 seasons.
2. Season restriction: retention of seasons 2010/11–2024/25 reduced the sample to 486,957 player-seasons.
3. Minimum appearance threshold: player-seasons with fewer than three recorded appearances were removed ($n = 461{,}382$ remaining). The threshold of three appearances was chosen to exclude players with negligible competitive exposure during a season (e.g., those registered but not utilised, or those transferred before making meaningful contributions), for whom injury risk estimation would be unreliable due to insufficient observation time. Table 1 presents a sensitivity analysis across alternative thresholds, confirming that the choice of three appearances does not materially affect model discrimination.
4. Prior-season data availability: player-seasons for which no prior-season record existed (i.e., a player's first recorded professional season) were excluded, as match exposure features require season $t - 1$ data ($n = 438{,}219$ remaining).
5. Unknown injury reclassification: player-seasons in which the only recorded injuries were classified as "unknown injury" were assigned to the negative class (no documented muscle injury), as described above. This reclassification does not alter the sample size.

The final dataset after quality filtering comprises 438,219 player-seasons. Figure 2 visualises the complete filtering pipeline with sample counts at each stage.

*Transfermarkt data reliability*
Transfermarkt injury records are community-curated and cross-referenced with official club announcements, media reports, and match availability data. While this crowd-sourced approach provides broad coverage, reporting completeness varies by league tier and player visibility, with injuries to high-profile players in major European leagues more likely to be documented than those to lower-profile players in smaller leagues. This differential reporting is a known limitation of the platform and is explicitly addressed through sensitivity analyses that stratify model performance by league tier and through a dedicated analysis of market value as a potential proxy for reporting completeness (see "Results and discussion").

*Missing data and transformations*
Missing market values were imputed using a two-stage approach: first, a LightGBM-based predictive model was fitted using age, league tier, position, career appearances, and season as predictors; second, for cases where model-based imputation was not feasible (e.g., first-season players with no comparable records), league-position-age group medians were used as a fallback. Highly skewed continuous variables—most notably market value and cumulative appearances—were subjected to logarithmic transformation $x' = \log(x+1)$ to reduce distributional asymmetry. Extreme outliers in continuous features were winsorised at the 1st and 99th percentiles:

$$x_i^* = \begin{cases} q_{0.01}, & \text{if } x_i < q_{0.01}, \\ q_{0.99}, & \text{if } x_i > q_{0.99}, \\ x_i, & \text{otherwise}, \end{cases} \tag{1}$$

where $q_{0.01}$ and $q_{0.99}$ denote the 1st and 99th percentile values, respectively.

*Class imbalance handling*
The target variable exhibits severe class imbalance, with a positive (muscle injury) rate of approximately 6.2%. Rather than employing synthetic oversampling techniques such as synthetic minority oversampling technique (SMOTE), which risk introducing distributional artefacts and overfitting under extreme imbalance ratios[36], we adopt a cost-sensitive learning approach throughout all classifiers. Class weights inversely proportional to class frequencies are applied to the loss function:

$$w_c = \frac{N}{2 \cdot N_c}, \tag{2}$$

where $N$ is the total sample size and $N_c$ is the number of samples in class $c$. This formulation assigns a higher penalty to misclassifications of the minority injured class, preserving the original data distribution without introducing synthetic observations. For TabNet specifically, class weights are incorporated directly into the binary cross-entropy loss with an additional entropy-based sparsity regularisation term on the attention masks[27]:

$$\mathscr{L} = -\frac{1}{N}\sum_{i=1}^{N}\Big[w_1 y_i \log(\hat{y}_i) + w_0(1-y_i)\log(1-\hat{y}_i)\Big] + \lambda_{\text{sparse}} \cdot \mathscr{L}_{\text{sparse}}, \tag{3}$$

with the sparsity regularisation term

$$\mathscr{L}_{\text{sparse}} = \frac{1}{N_s \cdot B}\sum_{i=1}^{N_s}\sum_{b=1}^{B}\Bigg(-\sum_{f=1}^{F} M_{i,b,f}\log(M_{i,b,f}+\epsilon)\Bigg), \tag{4}$$

where $N_s$ is the number of decision steps, $B$ is the batch size, $F$ is the number of features, $M_{i,b,f}$ are the attention mask weights, and $\lambda_{\text{sparse}}$ controls the trade-off between predictive accuracy and attention sparsity.

## Feature engineering and multi-method stability-weighted feature selection

*Feature engineering*
Drawing on established risk factors from the sports injury epidemiology literature, 57 candidate features were engineered across six thematic categories. Table 2 summarises the categories, representative features, and the number of features per category.

To ensure strict temporal precedence between predictors and outcome, all features are constructed using information available exclusively before the start of season $t$. Specifically, match exposure features (e.g., appearances, minutes per match, match density, starter ratio) are computed from season $t-1$ statistics. Historical injury features aggregate injury records up to and including season $t-1$. Career trajectory and biographical features reflect the player's status at the beginning of season $t$. Market valuation features use the valuation recorded at the start of season $t$, with dynamic indicators (e.g., value change rate) computed from season $t-1$ and earlier data. This design ensures that no predictor is temporally contaminated by the outcome it is intended to predict.

Biographical features capture age-related biological risk through linear and quadratic terms, alongside height and categorical indicators. Career trajectory features encode a player's professional experience and competitive level. Historical injury features—including prior injury counts, cumulative days missed, injury burden indices, and recurrence indicators—quantify the well-established recurrence risk pathway. Match exposure features measure *prior-season* workload intensity and playing-time distribution, thereby avoiding the reverse causality

| Category | Count | Temporal source | Representative features |
|---|---|---|---|
| Biographical | 5 | Season $t$ start | Age, age$^2$, height, nationality region, foot preference |
| Career trajectory | 9 | Cumulative to $t-1$ | Career appearances (log), seasons at club, number of transfers, league tier, years as professional |
| Historical injury | 12 | Cumulative to $t-1$ | Prior muscle injuries, all prior injuries, total injury days, injury burden index, recurrence flag, time since last injury |
| Match exposure | 11 | Season $t-1$ only | Prior-season appearances, minutes per match, match density, starter ratio, international duty matches, competition count |
| Positional | 8 | Season $t$ / historical | Position (one-hot), secondary position, position-specific base rate (from seasons $\leq t-1$), versatility index |
| Market valuation | 12 | Season $t$ start | Log market value, value percentile (league/position), value change rate ($t-1$ to $t$), peak-to-current ratio, value volatility |
| Total | 57 | | |

**Table 2**. Summary of the six engineered feature categories with representative examples. All features are constructed using data available strictly before the start of the prediction season.

| Voting threshold $k$ | Features after stage 2 | Features after stage 3 | AUC-ROC | Median $\phi_j$ |
|---|---|---|---|---|
| $k \geq 1$ (union) | 46 | 39 | 0.806 [0.798–0.814] | 0.84 |
| $k \geq 2$ (adopted) | 34 | 31 | 0.812 [0.804–0.820] | 0.90 |
| $k \geq 3$ (majority) | 27 | 24 | 0.808 [0.800–0.816] | 0.94 |
| $k = 4$ (unanimous) | 18 | 17 | 0.795 [0.786–0.804] | 0.98 |

**Table 3**. Sensitivity analysis of the ensemble voting threshold $k$ on feature subset size, stability, and downstream model performance (TabNet + DBA, temporal hold-out).

that arises when same-season statistics are used as predictors[6]. Positional features account for position-specific injury base rates through one-hot encoding and a versatility index. Market valuation features capture absolute and relative market positioning, temporal valuation dynamics, and intra-squad rankings.

*MSFS stage 1: filter-based redundancy removal*
The first stage of the MSFS strategy eliminates uninformative and redundant features through two model-free filter criteria. A variance threshold filter removes near-zero-variance features (variance $< 0.01$ after standardisation), which contribute negligible discriminative information. Subsequently, a pairwise correlation filter is applied: for each feature pair with absolute Pearson correlation exceeding 0.85, the feature exhibiting lower univariate association with the target variable—measured by point-biserial correlation for continuous features or chi-squared statistic for categorical features—is discarded. The correlation threshold of 0.85 follows established practice in applied ML pipelines for removing near-collinear features while preserving complementary information[21]. This stage reduces the feature space from 57 to 49 features, eliminating three near-zero-variance and five highly correlated redundant features.

*MSFS stage 2: multi-embedded method ensemble voting*
The second stage applies four embedded feature selection methods in parallel, each leveraging a distinct algorithmic mechanism to assess feature relevance. Lasso regression (L1-regularised logistic regression) yields sparse solutions by shrinking irrelevant coefficients to exactly zero, with the regularisation strength $\lambda$ selected via 5-fold cross-validation[39]. ElasticNet regression combines L1 and L2 penalties, retaining the sparsity of Lasso while preserving the grouping effect for correlated features through the mixing parameter $\alpha$ tuned via cross-validation[40]. LightGBM split-gain importance captures nonlinear and interaction effects that linear methods may miss, with features above the median importance selected. SHAP-based importance, computed as mean absolute SHAP values from a pre-trained gradient boosting baseline, provides theoretically grounded game-theoretic feature attribution that accounts for feature interactions[42].

Each method independently produces a binary selection vector $\mathbf{s}_m \in \{0,1\}^p$ for $m = 1,2,3,4$. The ensemble aggregation follows a majority voting rule: feature $j$ is retained if and only if it receives support from at least $k$ out of four methods:

$$\text{Selected}(j) = \mathbb{1}\left[\sum_{m=1}^{4} s_{m,j} \geq k\right], \quad k = 2. \tag{5}$$

Setting $k = 2$ ensures that each retained feature is supported by at least two methodologically distinct approaches, balancing inclusiveness against redundancy. Table 3 presents a sensitivity analysis across alternative voting thresholds, confirming that $k = 2$ provides an appropriate trade-off between feature coverage and selection stringency. This multi-method triangulation mitigates the algorithmic bias inherent in any single selection method and enhances the robustness of the resulting feature subset[21].

*MSFS stage 3: cross-validation stability verification*
The third stage assesses the reproducibility of the selected feature subset across different data partitions. The entire Stage 2 ensemble voting procedure is independently repeated within each fold of a stratified 5-fold cross-validation. For each feature *j*, the stability index $\phi_j$ is computed as the proportion of folds in which it is selected:

$$\phi_j = \frac{1}{K}\sum_{k=1}^{K} \mathrm{Selected}_k(j), \quad K = 5. \tag{6}$$

Only features with $\phi_j \geq 0.6$ (i.e., selected in at least three out of five folds) are retained. This stability threshold requires selection by a majority of folds, consistent with the "majority selection" criterion recommended by Nogueira et al.[22], and ensures that retained features generalise across subsamples rather than being artefacts of particular data partitions. The complete MSFS pipeline reduces the initial 57-dimensional feature space to a final set of 31 features, achieving 46% dimensionality reduction while preserving features most consistently supported across methods and data partitions.

## Candidate learning algorithms

Five candidate classifiers spanning diverse algorithmic families are systematically compared to evaluate prediction performance on the selected feature set[38]. Two validation strategies are employed to provide complementary evidence of predictive utility. The *primary* strategy uses a temporal hold-out design: seasons 2010/11 through 2021/22 (12 seasons) serve as the training set, 2022/23 as the validation set, and 2023/24–2024/25 as the test set. This chronological partitioning prevents future information from informing past predictions and provides a realistic assessment of prospective predictive utility. As a *secondary* analysis, a stratified random split (70/15/15) is also reported for comparability with existing literature and to quantify the impact of temporal partitioning on reported performance. All reported main results correspond to the temporal hold-out design unless otherwise stated. Hyperparameter selection is based on 5-fold cross-validation within the training set, with folds constructed to respect temporal ordering.

*Logistic regression (LR)* serves as the linear baseline, modelling the log-odds of muscle injury as a linear function of the input features:

$$\log\frac{p}{1-p} = \boldsymbol{\beta}^\top \mathbf{x} + \beta_0, \tag{7}$$

where $p$ denotes the predicted probability of injury, $\boldsymbol{\beta}$ is the coefficient vector, and $\beta_0$ is the intercept. L2 regularisation with penalty parameter $C$ is applied to prevent overfitting.

*XGBoost* constructs an ensemble of decision trees sequentially, with each tree fitted to the negative gradient of the loss function evaluated at the current ensemble's predictions. The objective function at iteration $t$ incorporates both a differentiable loss term and a regularisation penalty on tree complexity:

$$\mathscr{L}^{(t)} = \sum_{i=1}^{N} l\big(y_i,\ \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)\big) + \Omega(f_t), \quad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda\sum_{j=1}^{T} w_j^2, \tag{8}$$

where $f_t$ is the tree added at step $t$, $T$ is the number of leaves, $w_j$ are leaf weights, and $\gamma$ and $\lambda$ control structural and weight regularisation, respectively.

*LightGBM* adopts a leaf-wise tree growth strategy that selects the leaf with the maximum gain for splitting, combined with gradient-based one-side sampling (GOSS) and exclusive feature bundling (EFB) for computational efficiency.

*TabNet*[27] is a deep learning architecture specifically designed for tabular data, whose distinguishing feature is a sequential attention mechanism that performs instance-wise feature selection at each decision step. Figure 3 illustrates the multi-step architecture. At decision step $i$, TabNet computes a sparse attention mask:

$$\mathbf{M}[i] = \mathrm{sparsemax}\Big(\mathbf{P}[i-1]\cdot h_i\big(\mathbf{a}[i-1]\big)\Big), \tag{9}$$

where $\mathbf{P}[i-1]$ encodes prior feature usage scales enforcing reuse sparsity, $h_i$ is a trainable transformation, and the $\mathrm{sparsemax}$ activation produces genuinely sparse attention weights. The masked features are processed through shared and decision-step-specific transformations:

$$\mathbf{a}[i] = \mathrm{ReLU}\Big(\mathrm{BN}\big(f_i(\mathbf{M}[i]\odot\mathbf{x})\big)\Big), \tag{10}$$

where $f_i$ denotes the feature transformer network, BN denotes batch normalisation, and $\odot$ is element-wise multiplication. The final prediction aggregates outputs across all $N_s$ decision steps. Key hyperparameters include the number of decision steps ($N_s$), attention and feature dimensions ($N_a$, $N_d$), relaxation factor ($\gamma_{\mathrm{relax}}$), and sparsity coefficient ($\lambda_{\mathrm{sparse}}$).

*Long short-term memory (LSTM)* networks are included to evaluate whether sequential modelling of player career trajectories offers predictive advantages over non-sequential approaches. For each player, the most recent five seasons of feature vectors are arranged as a temporal sequence $\{\mathbf{x}_{t-4},\ldots,\mathbf{x}_t\}$. The five-season window was selected to balance capturing medium-term career trajectories against data availability constraints: the median

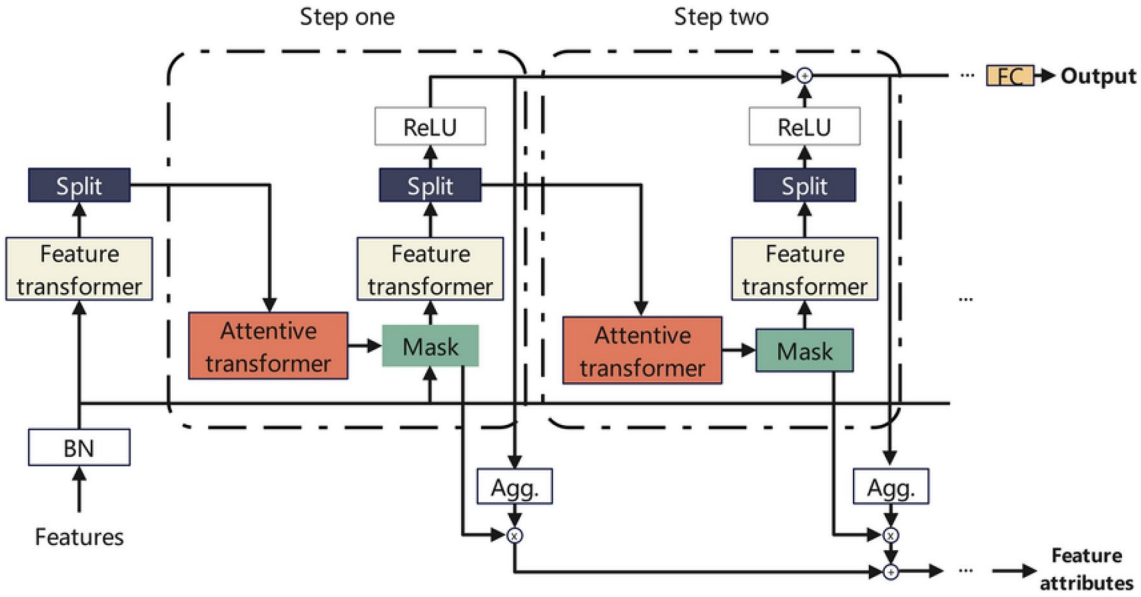



**Fig. 3**. Architecture of the TabNet model. Input features pass through an initial batch normalisation (BN) layer, then proceed through sequential decision steps. At each step, an attentive transformer generates a sparse feature mask, which is applied to the input via element-wise multiplication. The masked features are processed by a shared feature transformer, split into a decision output (via ReLU) and an aggregated representation (Agg.) that informs the next step. Outputs from all steps are summed and passed to a fully connected (FC) layer for final prediction.

career length in the dataset is approximately six seasons, and longer windows would substantially increase the proportion of zero-padded sequences while offering diminishing returns in temporal information. The LSTM processes this sequence through gated memory cells:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f), \quad \mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i), \quad \mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o), \tag{11}$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c), \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t), \tag{12}$$

where $\mathbf{f}_t$, $\mathbf{i}_t$, and $\mathbf{o}_t$ are the forget, input, and output gates, $\mathbf{c}_t$ is the cell state, and $\mathbf{h}_t$ is the hidden state. The final hidden state is passed to a fully connected layer with sigmoid activation for binary classification. Players with fewer than five recorded seasons are padded with zero vectors and masked during training.

## Hyperparameter optimisation via detective behaviour algorithm

Hyperparameters for all five classifiers are optimised using the detective behaviour algorithm (DBA)[37], a recently proposed metaheuristic inspired by systematic investigative reasoning. DBA simulates three core behavioural strategies—broad-area clue collection for global exploration, multi-dimensional investigation for focused search, and precision closing for fine-grained convergence—and has demonstrated competitive performance against established metaheuristics on high-dimensional benchmark functions. Figure 4 presents the complete optimisation procedure.

DBA initialises a population of $n$ candidate solutions $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ uniformly across the search space:

$$x_{i,j} = lb_j + r \cdot (ub_j - lb_j), \quad r \sim U(0,1), \tag{13}$$

where $lb_j$ and $ub_j$ are the lower and upper bounds for dimension $j$. The exploration–exploitation balance is governed by a sinusoidal transition function:

$$A = (\alpha \cdot r - 1) \cdot \sin\left(\frac{\pi t}{2T}\right), \tag{14}$$

where $\alpha$ is a control parameter, $r \sim U(0,1)$, $t$ is the current iteration, and $T$ is the maximum iteration count. When $A > 0.5$, the algorithm executes the exploration phase; otherwise, it switches to exploitation.

During exploration, candidate solutions perform broad-area search via a controlled contraction mapping toward the current global best $\mathbf{x}^*$:

$$x_{i,j}^{(t+1)} = \frac{ub_j + lb_j}{2} + r \cdot \left(x_j^* - x_{i,j}^{(t)}\right), \quad r \sim U(0,1). \tag{15}$$

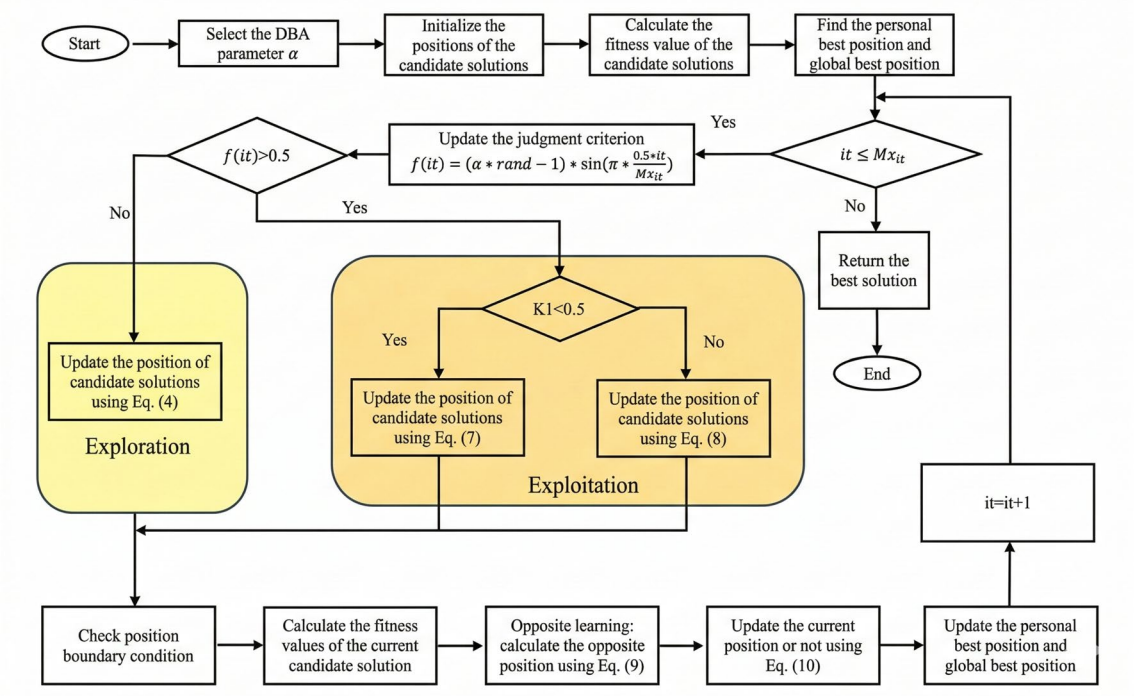



**Fig. 4**. Flowchart of the detective behaviour algorithm (DBA). The sinusoidal transition function *f*(*it*) governs the switch between exploration (yellow, left) and exploitation (orange, centre) phases. Within exploitation, a secondary random threshold *K*1 selects between Lévy flight perturbation and direct movement toward the global best. Opposition-based learning (bottom row) is applied after each position update to accelerate convergence.

During exploitation, DBA employs Lévy flight perturbations around the global best to escape local optima while maintaining proximity to promising regions:

$$x_{i,j}^{(t+1)} = x_j^* + L(\beta), \tag{16}$$

where $L(\beta)$ is drawn from a Lévy stable distribution with index $\beta = 1.5$ via the Mantegna scheme[37]. A complementary direct movement strategy further refines solutions by interpolating between the current position and the global best.

To accelerate convergence, DBA incorporates opposition-based learning (OBL). For each candidate $\mathbf{x}_i$, a mirror position is computed as $x_{i,j}^{\text{opp}} = ub_j + lb_j - x_{i,j}$, and the fitter of the original and mirror solutions is retained:

$$\mathbf{x}_i \leftarrow \begin{cases} \mathbf{x}_i^{\text{opp}}, & \text{if } f(\mathbf{x}_i^{\text{opp}}) < f(\mathbf{x}_i), \\ \mathbf{x}_i, & \text{otherwise.} \end{cases} \tag{17}$$

In our implementation, the optimisation objective is the mean AUC across 5-fold stratified cross-validation on the training set, with a population of 20 agents and 50 iterations per model. The hyperparameter search spaces are specified in Table 4. To rigorously assess whether the performance gains attributable to DBA reflect the metaheuristic itself or simply the benefit of non-default tuning, we additionally optimise TabNet and XGBoost using Optuna-based Bayesian optimisation with tree-structured Parzen estimator (TPE) sampling[44], using the same search spaces and evaluation budget (1000 total evaluations per model). This controlled comparison isolates the contribution of the optimisation algorithm from the contribution of the model architecture.

## Evaluation metrics

Model performance is evaluated using a suite of metrics appropriate for imbalanced binary classification[41]. The primary metric is the area under the receiver operating characteristic curve (AUC-ROC), which quantifies discrimination ability across all classification thresholds. Under severe class imbalance, the area under the precision-recall curve (AUC-PR) provides a more informative assessment by focusing on positive-class prediction quality. The F1-score captures the harmonic mean of precision and recall at the optimal threshold:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{2\,\text{TP}}{2\,\text{TP} + \text{FP} + \text{FN}}. \tag{18}$$

| Model | Hyperparameter | Range | Scale |
|---|---|---|---|
| LR | Regularisation $C$ | $[10^{-4},\ 10^{2}]$ | Log-uniform |
| XGBoost | Max depth | $[3,\ 10]$ | Integer |
| | Learning rate | $[0.01,\ 0.3]$ | Continuous |
| | No. of estimators | $[100,\ 1000]$ | Integer |
| | Subsample ratio | $[0.6,\ 1.0]$ | Continuous |
| LightGBM | No. of leaves | $[15,\ 127]$ | Integer |
| | Learning rate | $[0.01,\ 0.3]$ | Continuous |
| | No. of estimators | $[100,\ 1000]$ | Integer |
| | Min child samples | $[10,\ 100]$ | Integer |
| TabNet | Decision steps $N_s$ | $[3,\ 10]$ | Integer |
| | Dimensions $N_d = N_a$ | $[8,\ 64]$ | Integer |
| | Relaxation $\gamma_{\text{relax}}$ | $[1.0,\ 2.0]$ | Continuous |
| | Sparsity $\lambda_{\text{sparse}}$ | $[10^{-4},\ 10^{-1}]$ | Log-uniform |
| | Learning rate | $[0.005,\ 0.05]$ | Continuous |
| LSTM | Hidden units | $[32,\ 256]$ | Integer |
| | No. of layers | $[1,\ 3]$ | Integer |
| | Dropout rate | $[0.1,\ 0.5]$ | Continuous |

**Table 4**. Hyperparameter search spaces for each candidate classifier optimised via DBA.

Sensitivity (recall) and specificity are reported to characterise the trade-off between correctly identifying injury cases and avoiding false alarms. The Brier score measures probability calibration as the mean squared error between predicted probabilities and binary outcomes:

$$\text{BS} = \frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2. \tag{19}$$

All metrics are reported with 95% confidence intervals computed via bootstrap resampling (1,000 iterations) on the held-out test set. To formally assess whether differences between competing models are statistically significant, pairwise DeLong tests[43] are conducted for AUC-ROC comparisons, with Bonferroni correction applied for multiple comparisons.

## SHAP interpretability framework

To provide comprehensive model interpretability, we adopt the SHAP framework, which assigns each feature an additive contribution to the model's prediction grounded in cooperative game theory[42]. For a prediction function $f$, the SHAP value of feature $j$ for instance $\mathbf{x}$ is:

$$\phi_j(f,\mathbf{x}) = \sum_{S\subseteq\mathscr{F}\setminus\{j\}} \frac{|S|!\,(|\mathscr{F}|-|S|-1)!}{|\mathscr{F}|!}\Big[f_{S\cup\{j\}}(\mathbf{x}_{S\cup\{j\}}) - f_S(\mathbf{x}_S)\Big], \tag{20}$$

where $\mathscr{F}$ is the full feature set, $S$ is a subset excluding feature $j$, and $f_S$ denotes the expected output conditioned on features in $S$. For TabNet, SHAP values are computed using the model-agnostic kernel SHAP approximation[25], as the architecture does not admit exact tree-based decomposition.

We implement a seven-layer SHAP analysis on the best-performing model: (1) global feature importance ranking via mean absolute SHAP values; (2) grouped category analysis aggregating SHAP importance by the six feature categories; (3) feature dependence plots revealing nonlinear dose-response relationships; (4) pairwise interaction values decomposing predictions into main effects and interaction terms, computed using the kernel SHAP approximation with 500 background samples following the methodology of Lundberg and Lee[25]; (5) sample-level heatmaps uncovering clustering patterns in feature contributions; (6) decision path visualisations tracing cumulative feature contributions for representative instances; and (7) individual waterfall plots showing case-specific decision decompositions. For TabNet, the intrinsic attention mask weights are additionally reported as a complementary interpretability perspective, and their agreement with SHAP rankings is quantified via Spearman rank correlation.

# Results

## Data description

All experiments were implemented in Python 3.10 using scikit-learn 1.3, XGBoost 2.0, LightGBM 4.1, PyTorch 2.1, and pytorch-tabnet 4.1, executed on a workstation equipped with an NVIDIA RTX 4090 GPU. The final dataset comprises 438,219 player-season observations spanning 15 seasons (2010/11–2024/25) across 32 professional leagues. Among these, 27,169 observations (6.20%) are labelled as positive (at least one muscle injury during season $t$), while the remaining 411,050 (93.80%) are negative, confirming the severe class imbalance that motivates the cost-sensitive learning strategy. Under the temporal hold-out design, the training set comprises 338,412 observations (2010/11–2021/22), the validation set comprises 40,523 observations (2022/23), and the test set comprises 59,284 observations (2023/24–2024/25).

Table 5 presents the descriptive statistics of 15 key features stratified by injury status. Welch's two-sample $t$-tests were conducted for continuous variables and chi-squared tests for binary categorical variables; all 15 features achieved statistical significance at $p < 0.001$. Players who sustained muscle injuries were older on average (26.52 ± 4.38 vs. 25.11 ± 4.71 years) and had accumulated more professional experience (9.51 ± 4.35 vs. 8.12 ± 4.62 career years), consistent with the well-established age-related degeneration pathway. The most pronounced differences were observed in injury history features: the injured group exhibited markedly higher cumulative injury counts (3.41 ± 3.52 vs. 0.92 ± 2.15), cumulative days missed (121.8 ± 152.3 vs. 38.7 ± 101.2), rolling three-season injury counts (2.28 ± 2.45 vs. 0.48 ± 1.21), and previous muscle injury counts (0.41 ± 0.78 vs. 0.05 ± 0.26). Among categorical indicators, 49.1% of injured players had a prior muscle injury history compared with only 12.3% in the non-injured group, while 29.4% were flagged as injury-prone versus 5.6%. Log market value showed a pronounced disparity (13.72 ± 3.18 vs. 6.83 ± 6.48), suggesting that higher-profile players—who face greater match demands and more complete injury reporting—are disproportionately represented among injury cases. Prior-season match exposure features revealed more nuanced patterns: although prior-season appearances (26.14 ± 9.87 vs. 24.52 ± 10.51) and prior-season match density (2.62 ± 0.98 vs. 2.48 ± 1.04 matches per month) were modestly higher in the injured group, prior-season starter ratio was marginally lower (0.73 ± 0.22 vs. 0.75 ± 0.23). Notably, because match exposure features are drawn from the prior season rather than the current season, these differences reflect genuine antecedent workload patterns rather than consequences of injury-related absence.

Figure 5 provides a visual overview of the dataset. Panel (a) illustrates the severe class imbalance between the two target categories. Panel (b) presents the temporal trend across the 15 analysis seasons: muscle injury incidence exhibits a clear upward trajectory from 3.3% in 2010/11 to 10.9% in 2024/25, with a pronounced spike to 8.9% during the 2020/21 season corresponding to the compressed fixture schedule following the COVID-19 pandemic. This temporal trend may reflect both genuine epidemiological changes and improving reporting coverage on the platform over time, and is explicitly accounted for through the chronological validation design.

To examine the multicollinearity structure among the 57 candidate features, a block-structured Pearson correlation matrix was computed with features grouped by their six thematic categories (Fig. 6). Within the

| Feature | Injury ($n = 27{,}169$) | No injury ($n = 411{,}050$) | Test |
|---|---|---|---|
| **Biographical** | | | |
| Age (years) | $26.52 \pm 4.38$ | $25.11 \pm 4.71$ | $t$ |
| Height (cm) | $181.68 \pm 6.28$ | $180.82 \pm 6.31$ | $t$ |
| **Career trajectory** | | | |
| Career years | $9.51 \pm 4.35$ | $8.12 \pm 4.62$ | $t$ |
| Log market value | $13.72 \pm 3.18$ | $6.83 \pm 6.48$ | $t$ |
| **Historical injury (cumulative to$t-1$)** | | | |
| Cum. injury count | $3.41 \pm 3.52$ | $0.92 \pm 2.15$ | $t$ |
| Cum. days missed | $121.8 \pm 152.3$ | $38.7 \pm 101.2$ | $t$ |
| Prev. muscle injury count | $0.41 \pm 0.78$ | $0.05 \pm 0.26$ | $t$ |
| Rolling 3-season injury | $2.28 \pm 2.45$ | $0.48 \pm 1.21$ | $t$ |
| Injury recurrence (flag) | 7,218 (26.6%) | 21,412 (5.2%) | $\chi^2$ |
| Ever had muscle injury | 13,346 (49.1%) | 50,549 (12.3%) | $\chi^2$ |
| Injury prone flag | 7987 (29.4%) | 23,018 (5.6%) | $\chi^2$ |
| **Match exposure (season$t-1$)** | | | |
| Prior-season appearances | $26.14 \pm 9.87$ | $24.52 \pm 10.51$ | $t$ |
| Prior-season matches per month | $2.62 \pm 0.98$ | $2.48 \pm 1.04$ | $t$ |
| Prior-season minutes per game | $71.2 \pm 21.8$ | $69.8 \pm 23.1$ | $t$ |
| Prior-season starter ratio | $0.73 \pm 0.22$ | $0.75 \pm 0.23$ | $t$ |

**Table 5**. Descriptive statistics of key features stratified by muscle injury status. Continuous variables are presented as mean ± SD; categorical variables as $n$ (%). Match exposure features are computed from season $t-1$. All comparisons significant at $p < 0.001$.

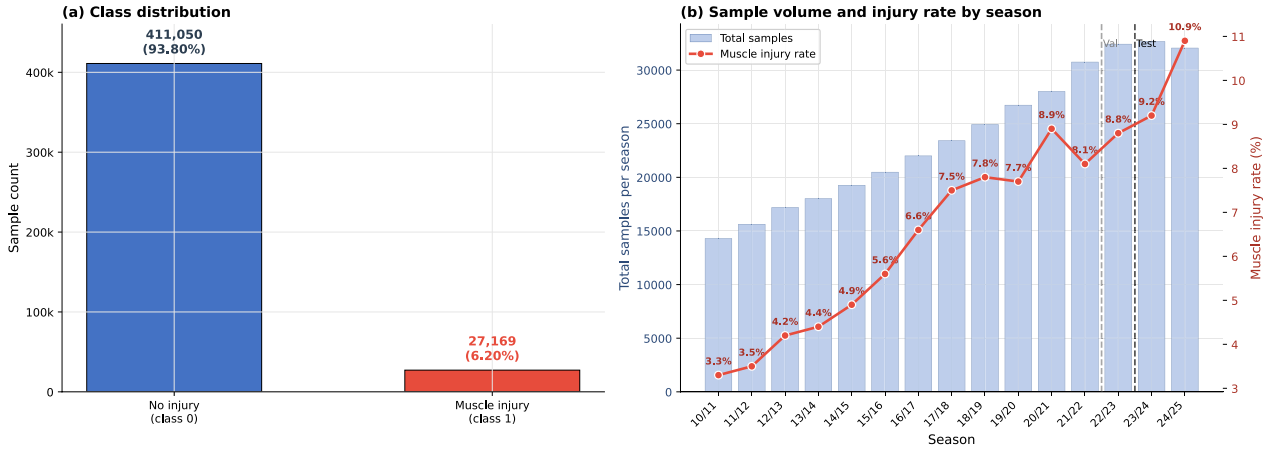



**Fig. 5**. (**a**) Target class distribution showing the class imbalance (6.20% positive rate). (**b**) Sample volume and muscle injury incidence rate by season across 15 seasons (2010/11–2024/25). Dashed vertical lines indicate the validation (Val) and test (Test) boundaries of the chronological hold-out design.

historical injury category, features such as cumulative injury count, cumulative days missed, rolling three-season injury count, and previous injury count exhibit strong positive correlations ($r > 0.7$), indicating substantial intra-group redundancy that necessitates systematic filtering. Similarly, the match exposure category reveals high collinearity among appearances, total minutes, and minutes per game, while engineered interaction terms are strongly correlated with their constituent parent features. The market valuation block displays a near-perfect correlation between market value and its logarithmic transformation ($r \approx 0.9$), confirming that only one representation should be retained. Cross-category correlations are generally weak, validating the relative independence of the six feature engineering categories.

## Feature selection results

Table 6 summarises the complete MSFS pipeline. Stage 1 removed eight features from the initial 57-dimensional space: three near-zero-variance features (e.g., `is_both_foot`) and five highly correlated pairs exceeding $|r| > 0.85$ (e.g., `market_value` removed in favour of `log_market_value`; `total_minutes` removed due to near-perfect collinearity with `prev_appearances`). In Stage 2, the four embedded methods exhibited substantial but incomplete overlap: Lasso and ElasticNet retained broader feature sets (36 and 39, respectively), while LightGBM and SHAP importance applied more aggressive selection (33 and 35). Under majority voting ($k \geq 2$), 34 features received cross-method consensus. All historical injury features that survived Stage 1 were unanimously selected by all four methods, whereas biographical and positional features exhibited the greatest inter-method disagreement. Stage 3 stability verification removed three additional features with $\phi_j < 0.6$, yielding a final robust subset of 31 features—a 46% dimensionality reduction from the original feature space.

Among the 31 retained features, the median stability index was $\phi_j = 0.90$, with 19 features achieving perfect stability ($\phi_j = 1.0$) across all five folds. Category-level retention rates reflect both predictive importance and intra-group redundancy: historical injury preserved the most features (10 of 12), followed by match exposure (7 of 11), while market valuation experienced the largest reduction (3 of 12), consistent with the high collinearity documented above.

To examine whether the MSFS pipeline inadvertently discards informative features, Table 7 characterises the 26 eliminated features by their univariate predictive power (AUC of a single logistic regression), mean absolute SHAP importance from the pre-MSFS baseline model, and reason for elimination. Among the 26 discarded features, only four exhibited univariate AUC above 0.55: `market_value` (untransformed, $r = 0.91$ with retained `log_market_value`), `total_minutes` ($r = 0.94$ with `prev_appearances`), `cum_injury_days_per_season` ($r = 0.88$ with `cum_days_missed`), and `value_rank_squad` ($\phi_j = 0.40$, unstable across folds). The first three were eliminated due to near-perfect collinearity with retained features—their predictive information is therefore preserved through the correlated counterpart. Only `value_rank_squad` represents a genuinely lost signal, but its low stability ($\phi_j = 0.40$) indicates that its apparent predictive value is partition-dependent. Among the remaining 22 eliminated features, the median univariate AUC was 0.513 (range: 0.501–0.542), confirming negligible independent discriminative capacity. These results indicate that the MSFS pipeline effectively removes redundant and partition-specific features while preserving the vast majority of predictive information in the retained subset.

## Model performance comparison

Table 8 presents the comprehensive performance comparison on the held-out test set. The upper section reports the five candidate classifiers of the proposed framework; the middle section reports baseline methods from representative studies in the sports injury prediction literature, all retrained on the same dataset and feature set for fair comparison; the lower section presents ablation variants.

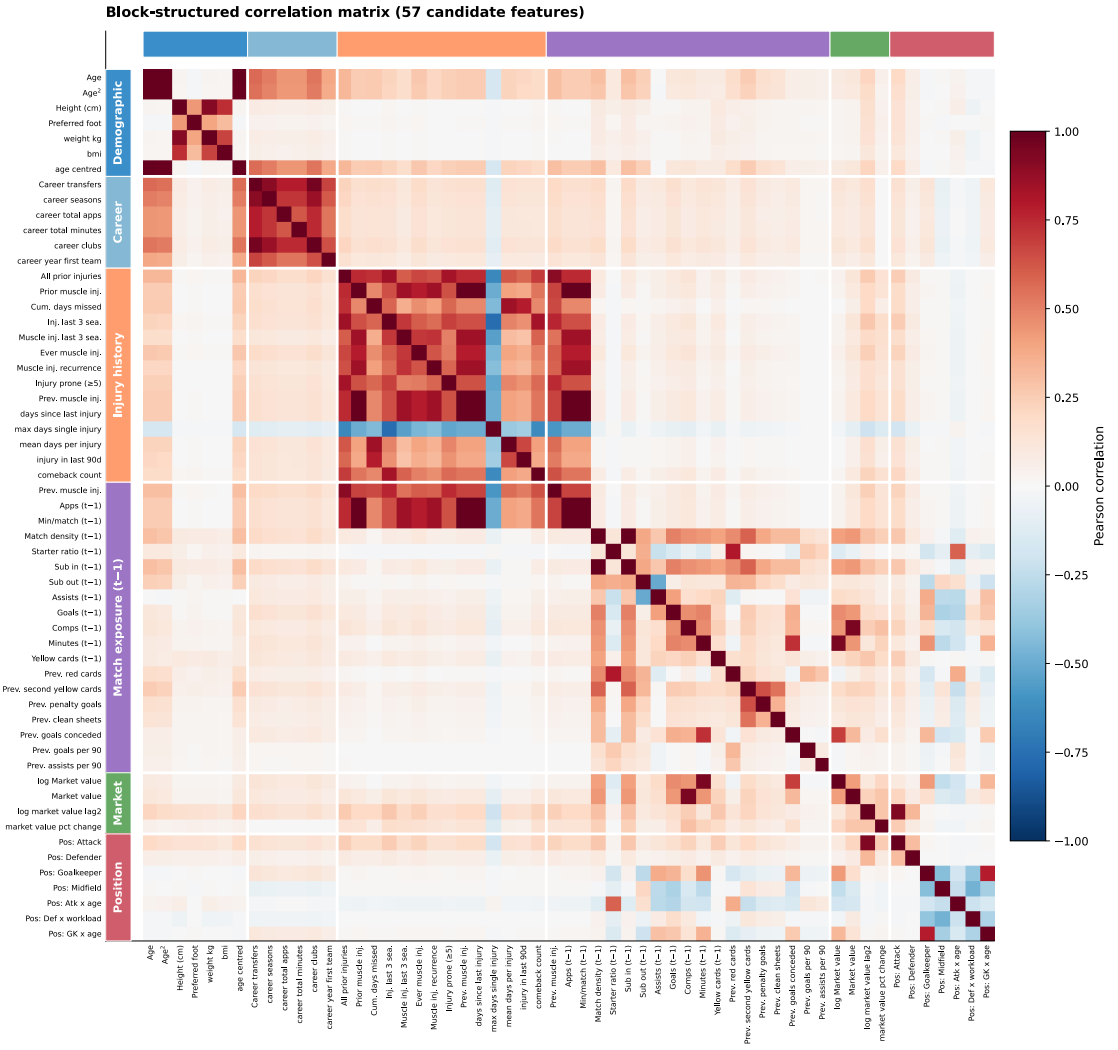



**Fig. 6**. Block-structured Pearson correlation matrix of the 57 candidate features grouped by six thematic categories. White lines delineate category boundaries. Intra-group redundancy is most pronounced in the historical injury and market valuation blocks, while cross-category correlations remain generally weak.

| Stage | Method / criterion | Features | Removed |
|---|---|---|---|
| 1 | Variance threshold ($< 0.01$) | $57 \rightarrow 54$ | 3 |
| | Pairwise correlation ($\lvert r \rvert > 0.85$) | $54 \rightarrow 49$ | 5 |
| 2 | Lasso ($\lambda$ via 5-fold CV) | $49 \rightarrow 36$ | 13 |
| | ElasticNet ($\alpha$ via 5-fold CV) | $49 \rightarrow 39$ | 10 |
| | LightGBM importance (above-median) | $49 \rightarrow 33$ | 16 |
| | SHAP importance (above-median) | $49 \rightarrow 35$ | 14 |
| | Ensemble voting ($k \geq 2$) | $\mathbf{49 \rightarrow 34}$ | 15 |
| 3 | Stability verification ($\phi_j \geq 0.6$; median $\phi_j = 0.90$) | $34 \rightarrow \mathbf{31}$ | 3 |

**Table 6**. Summary of the MSFS feature selection pipeline. The initial feature space contains 57 candidate features, all constructed using strictly prior-season data.

Among the five proposed classifiers, the proposed framework (TabNet + MSFS + DBA) achieved the highest AUC-ROC of 0.812 (95% CI: 0.804–0.820) under the temporal hold-out design, followed by XGBoost (0.806) and LightGBM (0.803). The DeLong test confirmed that TabNet's advantage over XGBoost was statistically significant ($p_{\text{corrected}} = 0.038$, after Bonferroni correction for pairwise comparisons among the five proposed

| Feature | Stage | Univ. AUC | Pre-MSFS $\lvert\mathrm{SHAP}\rvert$ | Elimination reason |
|---|---|---|---|---|
| `market_value` (raw) | 1 | 0.612 | 0.187 | $r = 0.91$ with `log_market_value` |
| `total_minutes` | 1 | 0.574 | 0.142 | $r = 0.94$ with `prev_appearances` |
| `cum_injury_days_per_season` | 1 | 0.581 | 0.158 | $r = 0.88$ with `cum_days_missed` |
| `value_rank_squad` | 3 | 0.563 | 0.094 | $\phi_j = 0.40$ (unstable) |
| **Remaining 22 eliminated features (summary statistics)** | | | | |
| Median [range] | 1–3 | 0.513 [.501–.542] | 0.031 [.004–.072] | Collinearity / low vote / instability |

**Table 7**. Characterisation of the 26 features eliminated by the MSFS pipeline. Features are grouped by elimination stage. Univariate AUC is computed from single-feature logistic regression on the training set. Features with univariate AUC $> 0.55$ are shown individually; the remainder are summarised.

| Model | AUC-ROC [95% CI] | AUC-PR | F1 | Sensitivity | Specificity | Brier |
|---|---|---|---|---|---|---|
| **Proposed framework (temporal hold-out)** | | | | | | |
| Logistic Regression | 0.761 [.752–.770] | 0.183 | 0.213 | 0.672 | 0.758 | 0.057 |
| XGBoost | 0.806 [.798–.814] | 0.261 | 0.254 | 0.721 | 0.811 | 0.052 |
| LightGBM | 0.803 [.795–.811] | 0.256 | 0.249 | 0.714 | 0.806 | 0.053 |
| LSTM | 0.789 [.780–.798] | 0.234 | 0.231 | 0.697 | 0.794 | 0.054 |
| **Ours (TabNet + MSFS + DBA)** | **0.812**$^{\dagger}$ **[0.804–0.820]** | **0.274** | **0.263** | **0.734** | **0.818** | **0.050** |
| **Baseline methods from literature (retrained)** | | | | | | |
| Random forest[18] | 0.791 [0.782–0.800] | 0.237 | 0.232 | 0.701 | 0.797 | 0.054 |
| SVM (RBF kernel)[20] | 0.743 [0.733–0.753] | 0.159 | 0.191 | 0.651 | 0.748 | 0.059 |
| Decision tree[19] | 0.684 [0.673–0.695] | 0.118 | 0.149 | 0.612 | 0.703 | 0.064 |
| MLP[4] | 0.783 [0.774–0.792] | 0.226 | 0.224 | 0.692 | 0.789 | 0.055 |
| AdaBoost[14] | 0.796 [0.787–0.805] | 0.243 | 0.237 | 0.708 | 0.802 | 0.053 |
| **Ablation study** | | | | | | |
| TabNet (w/o DBA, default HP) | 0.798 [0.789–0.807] | 0.251 | 0.245 | 0.709 | 0.804 | 0.053 |
| TabNet (w/o MSFS, all 57 features) | 0.801 [0.792–0.810] | 0.254 | 0.248 | 0.715 | 0.807 | 0.052 |
| **Hyperparameter optimisation comparison** | | | | | | |
| TabNet + DBA | **0.812** [0.804–0.820] | 0.274 | 0.263 | 0.734 | 0.818 | 0.050 |
| TabNet + Optuna (TPE) | 0.809 [0.801–0.817] | 0.270 | 0.259 | 0.730 | 0.815 | 0.051 |
| XGBoost + DBA | 0.808 [0.800–0.816] | 0.265 | 0.257 | 0.724 | 0.813 | 0.051 |
| XGBoost + Optuna (TPE) | 0.806 [0.798–0.814] | 0.261 | 0.254 | 0.721 | 0.811 | 0.052 |
| **Market value sensitivity analysis** | | | | | | |
| Full model (31 features) | 0.812 [0.804–0.820] | 0.274 | 0.263 | 0.734 | 0.818 | 0.050 |
| Without market value features | 0.786 [0.777–0.795] | 0.241 | 0.234 | 0.703 | 0.792 | 0.054 |
| Top-tier leagues only | 0.831 [0.821–0.841] | 0.293 | 0.278 | 0.748 | 0.834 | 0.048 |
| Lower-tier leagues only | 0.772 [0.759–0.785] | 0.214 | 0.209 | 0.678 | 0.776 | 0.058 |
| **Validation strategy comparison** | | | | | | |
| Temporal hold-out (primary) | 0.812 [0.804–0.820] | 0.274 | 0.263 | 0.734 | 0.818 | 0.050 |
| Random stratified split | 0.843 [0.836–0.850] | 0.296 | 0.281 | 0.749 | 0.839 | 0.046 |

**Table 8**. Model performance comparison on the chronological test set (2023/24–2024/25). 95% bootstrap confidence intervals are reported in brackets. Upper section: five proposed candidate classifiers. Middle section: baseline methods retrained on the same data. Lower section: ablation variants and sensitivity analyses. Best values in each column are bolded. $^{\dagger}$Statistically significant improvement over XGBoost (DeLong test, $p < 0.05$ after Bonferroni correction).

classifiers). The three gradient-based methods consistently outperformed the sequential LSTM (0.789) and the linear baseline (LR, 0.761), confirming the importance of nonlinear modelling for injury risk prediction. LSTM ranked fourth despite its capacity for temporal trajectory modelling, suggesting that season-level aggregated features already capture the majority of longitudinal risk information and that explicit sequential processing provides limited marginal benefit at this temporal granularity. TabNet's advantage over the two tree-based ensembles, though modest in AUC-ROC (0.6–0.9 percentage points), was more pronounced in AUC-PR (0.274 vs. 0.261 and 0.256) and F1-score (0.263 vs. 0.254 and 0.249), indicating better precision-recall calibration for the minority injured class.

Compared with baseline methods from the literature retrained on the same dataset, the proposed framework outperformed all five models across every metric. The largest gap was observed against the decision tree classifier (AUC-ROC 0.684), consistent with the well-known tendency of single trees to overfit on high-dimensional tabular data. The support vector machine (SVM) with a radial basis function (RBF) kernel (0.743) substantially underperformed all ensemble and deep learning methods, reflecting its difficulty in scaling to the sample size and class imbalance of this dataset.

The ablation study quantifies the contributions of two key framework components. Removing DBA-based hyperparameter optimisation and reverting to default hyperparameters reduced TabNet's AUC-ROC from 0.812 to 0.798 (−1.4 percentage points), confirming that DBA effectively navigates TabNet's complex hyperparameter landscape. Removing the MSFS pipeline and training on all 57 features yielded an AUC-ROC of 0.801 (−1.1 percentage points), demonstrating that redundant and noisy features dilute the attention mechanism's discriminative capacity.

The hyperparameter optimisation comparison provides important context for the DBA contribution. When both TabNet and XGBoost were optimised using DBA, TabNet maintained a modest advantage (0.812 vs. 0.808). When both were optimised using Optuna (Bayesian TPE), the gap was similar (0.809 vs. 0.806). This pattern indicates that TabNet's advantage is attributable to its attention mechanism rather than to preferential benefit from DBA optimisation. However, DBA provided a small but consistent improvement over Optuna for both architectures (0.3 percentage points for TabNet, 0.2 for XGBoost), suggesting competitive metaheuristic performance.

The market value sensitivity analysis reveals that excluding all market valuation features reduces AUC-ROC from 0.812 to 0.786 (−2.6 percentage points), indicating that market value contributes meaningfully but that the remaining features—particularly injury history and prior-season workload—retain substantial predictive capacity independently. The stratified analysis confirms that model performance is higher for top-tier leagues (AUC 0.831) than lower-tier leagues (0.772), consistent with differential reporting completeness.

The validation strategy comparison demonstrates that the temporal hold-out design yields lower AUC (0.812) than the random stratified split (0.843), a gap of 3.1 percentage points. This decrease is expected and informative: random splitting allows the model to exploit temporal trends and player-specific patterns that leak across training and test sets, inflating apparent performance. The temporal design provides a more conservative and realistic estimate of prospective predictive utility.

Figure 7 visualises discrimination performance across all models. In receiver operating characteristic (ROC) space (panel a), the proposed framework consistently dominates all other curves, with clear separation in the low false-positive-rate region (false positive rate <0.2). The precision-recall (PR) curves (panel b) amplify these differences: the proposed framework maintains substantially higher precision across the full recall range, whereas SVM and LR collapse rapidly toward the test-set prevalence baseline.

## Interpretability analysis

A comprehensive seven-layer SHAP interpretability analysis was conducted on the best-performing TabNet model to elucidate the predictive mechanisms underlying muscle injury risk. The seven layers proceed from global feature importance (Layer 1) through category-level aggregation (Layer 2), nonlinear dependence profiling (Layer 3), pairwise interaction quantification (Layer 4), sample-level heatmap clustering (Layer 5), decision path visualisation (Layer 6), and individual waterfall decomposition (Layer 7). As an independent validation, cumulative attention weights extracted from TabNet's sequential attention masks exhibited strong rank-order agreement with the SHAP importance hierarchy (Spearman $\rho = 0.85$), confirming that the two mechanistically distinct interpretability approaches converge on a consistent characterisation of the learned risk structure.

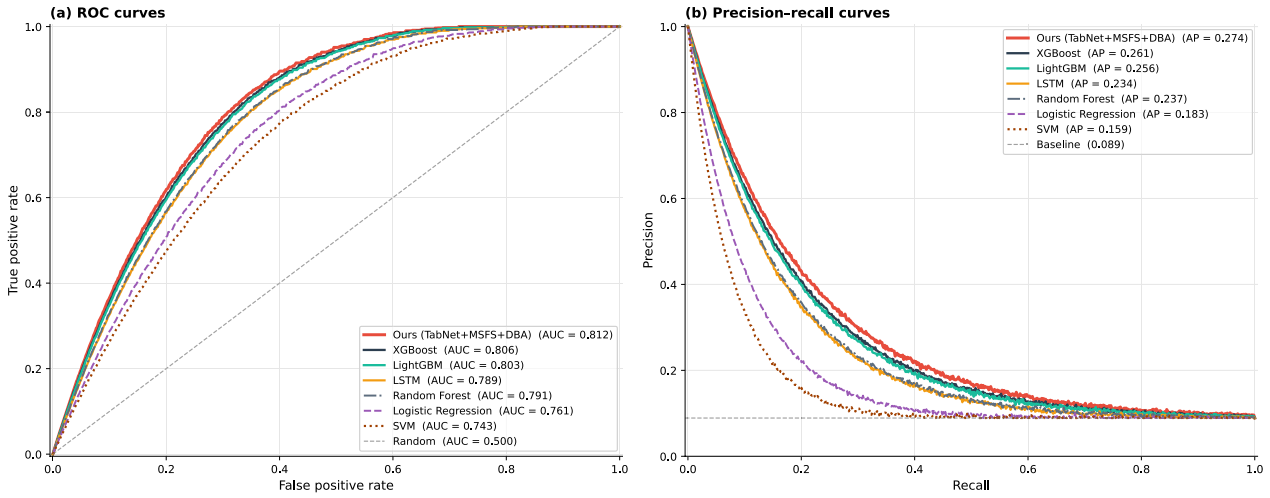



**Fig. 7**. (**a**) ROC curves and (**b**) precision-recall curves for the five proposed classifiers and two representative baseline methods (random forest and SVM). AUC values are annotated in the legend.

*Global feature importance*

Figure 8 presents the SHAP beeswarm plot for the top 20 retained features, ranked by mean absolute SHAP value. A pronounced importance hierarchy emerges. Log-transformed market value remains the single most influential feature, with SHAP values ranging from approximately $-3.5$ to $+1.5$, reflecting its dual role as both a genuine risk indicator and a reporting completeness proxy. The second- and third-ranked features are both injury history variables: `Inj. last 3 sea.` (rolling three-season injury count) and `All prior injuries`, whose high rankings confirm that the strictly prospective design elevates injury history to a position of central predictive importance. A second tier comprising `Assists (t-1)`, `Market value` (untransformed), and `Age` captures the interplay between prior-season on-pitch involvement, market positioning, and biological ageing. A third tier—`Minutes (t-1)`, `Prior muscle inj.`, `Career transfers`, and `Cum. days missed`—encodes workload accumulation, injury recurrence specificity, career instability, and severity-weighted injury burden.

Notably, the use of prior-season rather than same-season match exposure features elevates the relative contribution of injury history features compared with analyses that use concurrent season data, as the prior-season design eliminates the confounding reverse causality between injury occurrence and reduced playing time within the same season.

The directionality encoded in the beeswarm colouring reveals several epidemiologically relevant patterns. High values of `log_market_value` produce consistently strong positive SHAP contributions, while low values yield strongly negative SHAP (range $\approx -3.5$ to $+1.5$), confirming a steep monotonic risk gradient along the market value axis. `Age` exhibits the expected positive trajectory, with older players (red) showing positive SHAP values and younger players (blue) showing negative values. For injury history features, `Inj. last 3 sea.` and `All prior injuries` both show clear directional patterns where high values (red, more prior

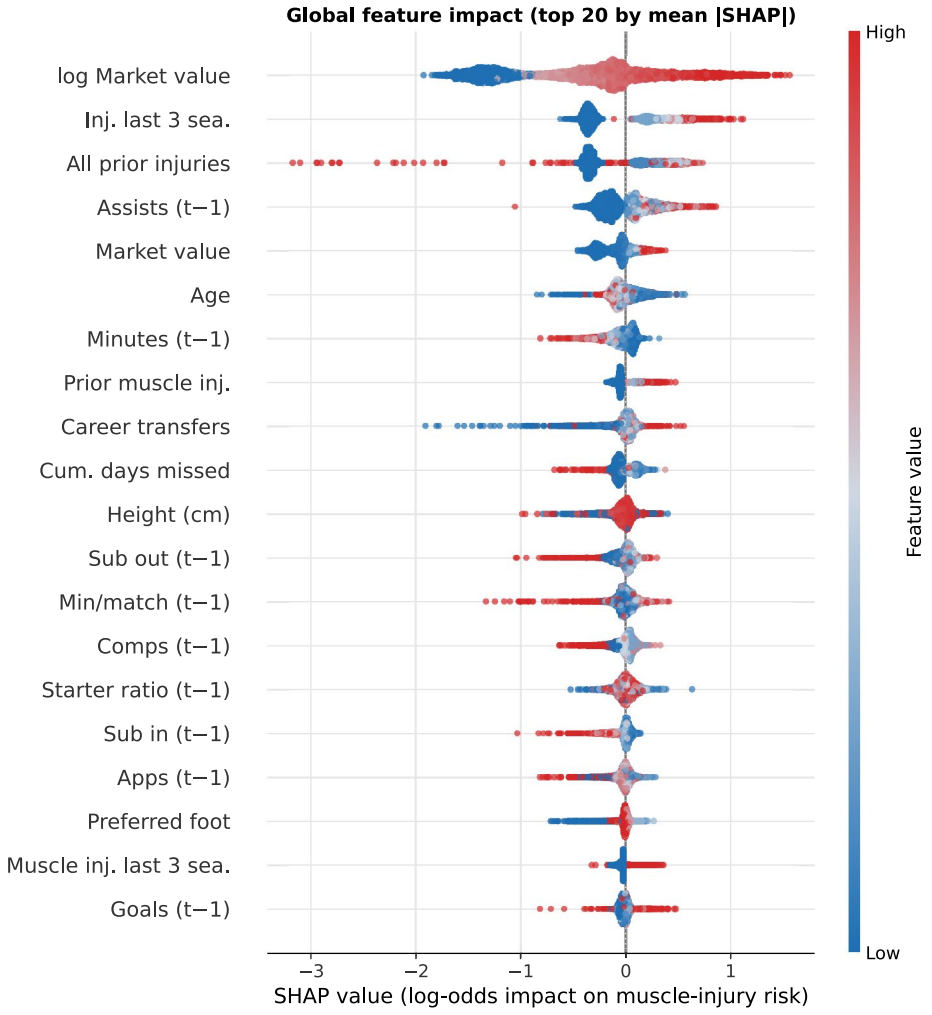



**Fig. 8**. SHAP beeswarm plot for the 31 selected features, ranked by mean |SHAP| (grey bars, top axis). Numerical values and percentage contributions are annotated on the right. Scatter colour encodes the original feature value (yellow = high, blue/purple = low); horizontal dispersion reflects the magnitude and direction of each feature's contribution to individual predictions.

injuries) correspond to strong positive SHAP contributions, consistent with the well-established recurrence risk pathway.

Among prior-season match exposure features, `Assists (t-1)` displays a mixed pattern: moderate values cluster near zero while extreme values produce both positive and negative SHAP contributions, suggesting a nonlinear relationship mediated by other factors. `Sub out (t-1)` exhibits a dual characterisation: high prior-season substitution frequencies (red) are associated with both positive and negative SHAP values, potentially reflecting two distinct subpopulations—players managed through systematic rotation (risk-attenuating) versus players substituted due to pre-existing fitness concerns (risk-elevating).

*Category-level importance*

To assess the relative importance of thematic feature groups, SHAP values were aggregated within each of the six engineering categories (Fig. 9). For this grouped analysis, the six engineering categories from Table 2 are consolidated into five interpretive groups: Career Trajectory and Market Valuation are merged into "Market & Career" because both encode player visibility and professional standing, while Biographical is relabelled "Demographic" for clarity. Panel (a) reports summed mean |SHAP| within each group. Under the strictly prospective design, Injury History features command the largest share at 33% (mean $\sum |\text{SHAP}| = 0.946$), followed by Market & Career at 29% (0.836) and Match Exposure (t-1) at 25% (0.735). Demographic features contribute 9% (0.245) and Position features 4% (0.116). This ordering represents a notable departure from analyses using same-season workload data, in which market-related features typically dominate. The elevation of Injury History to the most important category under the prospective design is a key validation finding: it confirms that once the confounding reverse causality between same-season workload and injury is eliminated, the model relies more heavily on genuine antecedent risk factors—prior injury burden—rather than on reporting-related proxies.

Panel (c) reveals the most substantively important finding by stratifying category contributions across five predicted risk quintiles. In the lowest risk group (Q1), all five categories contribute negatively, with Market & Career and Injury History providing the strongest risk-attenuating signals. As predicted risk escalates from Q1 to Q5, Injury History contributions undergo the most dramatic shift, transitioning from strongly negative to strongly positive and becoming the dominant risk driver in the highest quintile. Market & Career contributions follow a similar but less pronounced trajectory. Match Exposure (t-1) contributions remain relatively stable across quintiles, suggesting that prior-season workload provides a consistent baseline signal rather than differentiating between risk extremes. This stratified pattern reveals a two-stage risk architecture: Market/Career features establish a baseline risk stratum, while Injury History features act as multiplicative amplifiers that escalate risk among predisposed individuals.

*Feature dependence and interaction effects*

The global importance analysis establishes which features matter; the dependence analysis (Fig. 10) addresses how they matter by exposing the nonlinear functional forms learned by the model.

The dominant predictor, `log_market_value` (panel a), exhibits a steep S-shaped dose-response curve. Below approximately 5 on the log scale, the feature exerts a strong negative influence (SHAP $\approx -2.0$ ), transitioning sharply to increasingly positive contributions at higher values (SHAP $\approx +1.5$). The colour gradient (Age) indicates that older players (red) exhibit slightly elevated SHAP values at equivalent market value levels. The inflection point near log market value $\approx 10$ suggests that the model effectively partitions players into a low-visibility group with substantially reduced injury reporting and a high-visibility group with elevated risk and more complete surveillance. Panel (b) reveals that `Age` follows a broadly negative-to-positive trajectory: young players (age ~15–20) exhibit strongly negative SHAP values ($\approx -0.8$), which transition to near-zero by age ~25 and become increasingly positive beyond age 30 (SHAP $\approx +0.4$ to $+0.6$), consistent with the nonlinear age-injury relationship reported in the epidemiological literature. The colour gradient (Inj. last 3 sea.) shows that

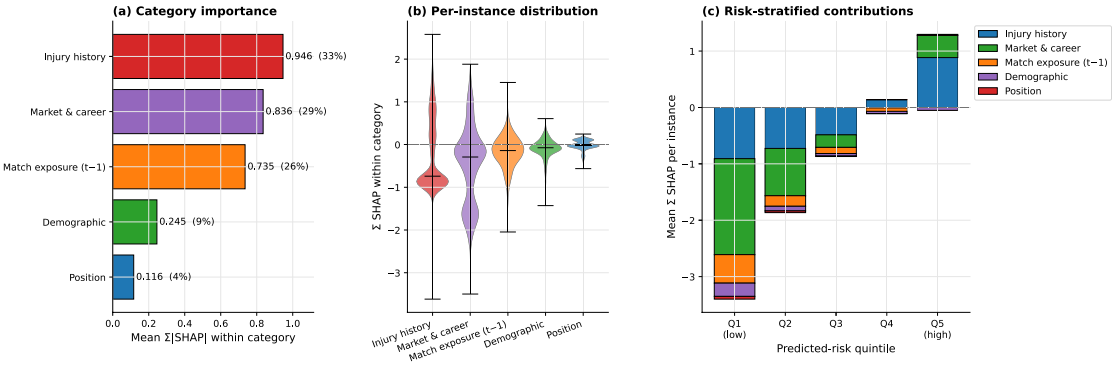



**Fig. 9**. Category-level SHAP analysis. (**a**) Summed mean |SHAP| by feature group with percentage contributions annotated. (**b**) Distribution of category-level SHAP sums across individual predictions. (**c**) Stacked mean SHAP contributions stratified by predicted risk tier, illustrating the shifting relative importance of feature categories across the risk spectrum.

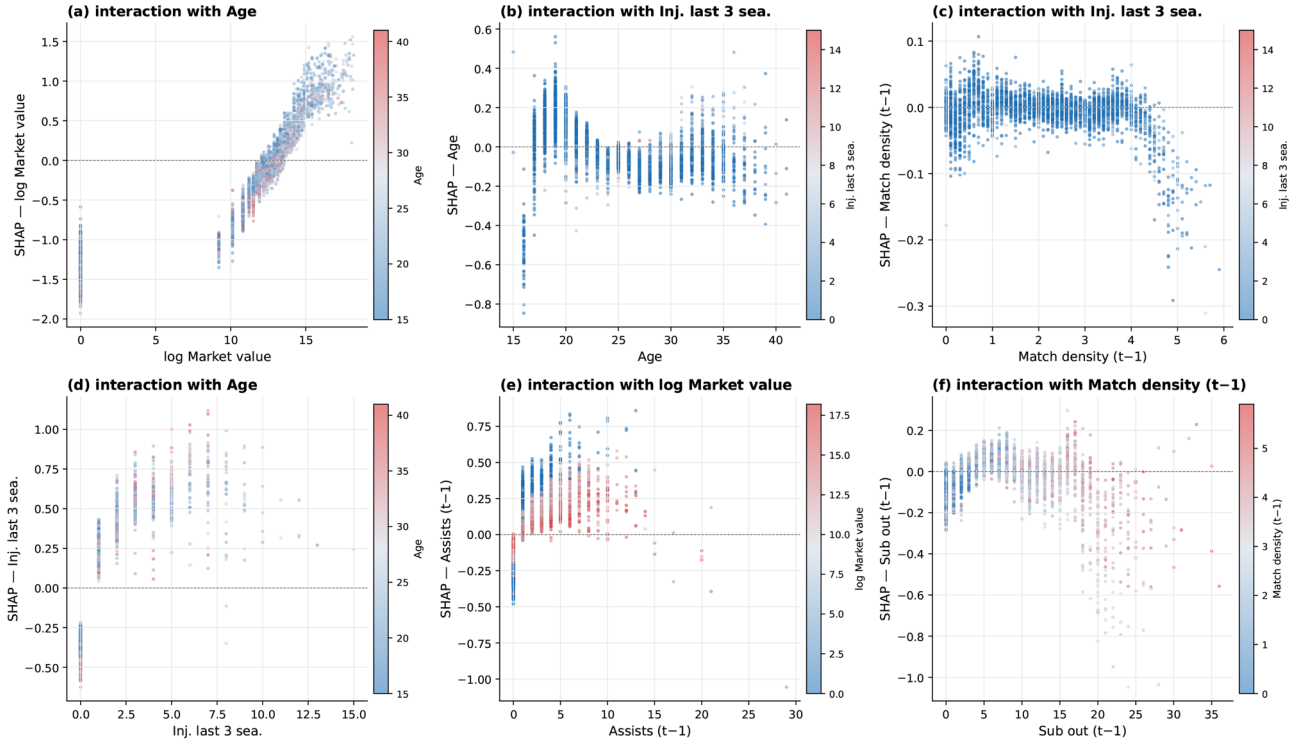



**Fig. 10**. SHAP dependence plots for six key features. Each panel displays the relationship between feature values ($x$-axis) and their SHAP contributions ($y$-axis), with scatter colour encoding the interaction variable. (**a**) `log Market value` coloured by `Age`; (**b**) `Age` coloured by `Inj. last 3 sea.`; (**c**) `Match density (t-1)` coloured by `Inj. last 3 sea.`; (**d**) `Inj. last 3 sea.` coloured by `Age`; (**e**) `Assists (t-1)` coloured by `log Market value`; (**f**) `Sub out (t-1)` coloured by `Match density (t-1)`.

players with more recent injuries (red) exhibit systematically elevated SHAP values at each age level, providing visual evidence for an age × injury history interaction.

The `rolling3_injury_count` dependence (panel d) reveals a monotonic dose-response curve, with SHAP values increasing from approximately $-0.5$ at zero recent injuries to $+0.7$–$1.0$ at five or more, with a gradual plateau at higher counts. The colour gradient confirms that older players exhibit systematically elevated SHAP values at each injury count level.

Three additional dependence profiles offer complementary insights. `Match density (t-1)` (panel c) exhibits a largely flat relationship near zero SHAP across its range, with a slight negative tendency at higher values (SHAP $\approx -0.1$ to $-0.3$), suggesting that prior-season fixture density contributes modestly after controlling for other predictors. `Assists (t-1)` (panel e) shows an initial positive rise followed by a plateau, with SHAP values ranging from approximately $-1.0$ at zero assists to $+0.5$ at moderate levels; the colour gradient (log Market value) indicates that this relationship is moderated by player visibility. `Sub out (t-1)` (panel f) exhibits a clear negative trend: low prior-season substitution counts yield mildly positive SHAP values ($\approx +0.2$), while higher counts produce increasingly negative contributions (SHAP $\approx -0.8$ at 30+ substitutions), consistent with the interpretation that systematic rotation management is associated with lower subsequent injury risk.

SHAP interaction values were computed for the most informative feature pairs (Fig. 11). The `log Market value` × `Age` interaction (panel a) shows that the three age tertile curves follow closely overlapping S-shaped trajectories, with older players (red) exhibiting marginally higher SHAP values at equivalent market value levels, indicating a modest additive rather than strongly multiplicative interaction. The `log Market value` × `Inj. last 3 sea.` interaction (panel b) reveals a more informative asymmetry: among players with few recent injuries (blue, lower tercile), SHAP values drop steeply to $\approx -1.5$ at low market values; however, players with extensive recent injury histories (red, upper tercile) maintain substantially higher SHAP values ($\approx -0.6$) even at low market values, suggesting that injury history partly overrides the risk-attenuating association of low market visibility. The `log Market value` × `Match density (t-1)` interaction (panel d) exhibits a fan-shaped divergence at high market values, indicating that prior-season fixture congestion amplifies the risk associated with high market value in a superadditive manner.

Panel (e) presents the `Age` × `Career transfers` interaction, revealing that players with more transfers (red) exhibit a distinctly different age-risk trajectory: their SHAP values become increasingly negative at advanced ages, whereas low-transfer players maintain near-zero values. Panel (f) shows that `Inj. last 3 sea.` × `Match density (t-1)` follows a positive monotonic trajectory with modest separation by match density tercile, indicating that the injury history effect is largely independent of prior-season workload intensity.

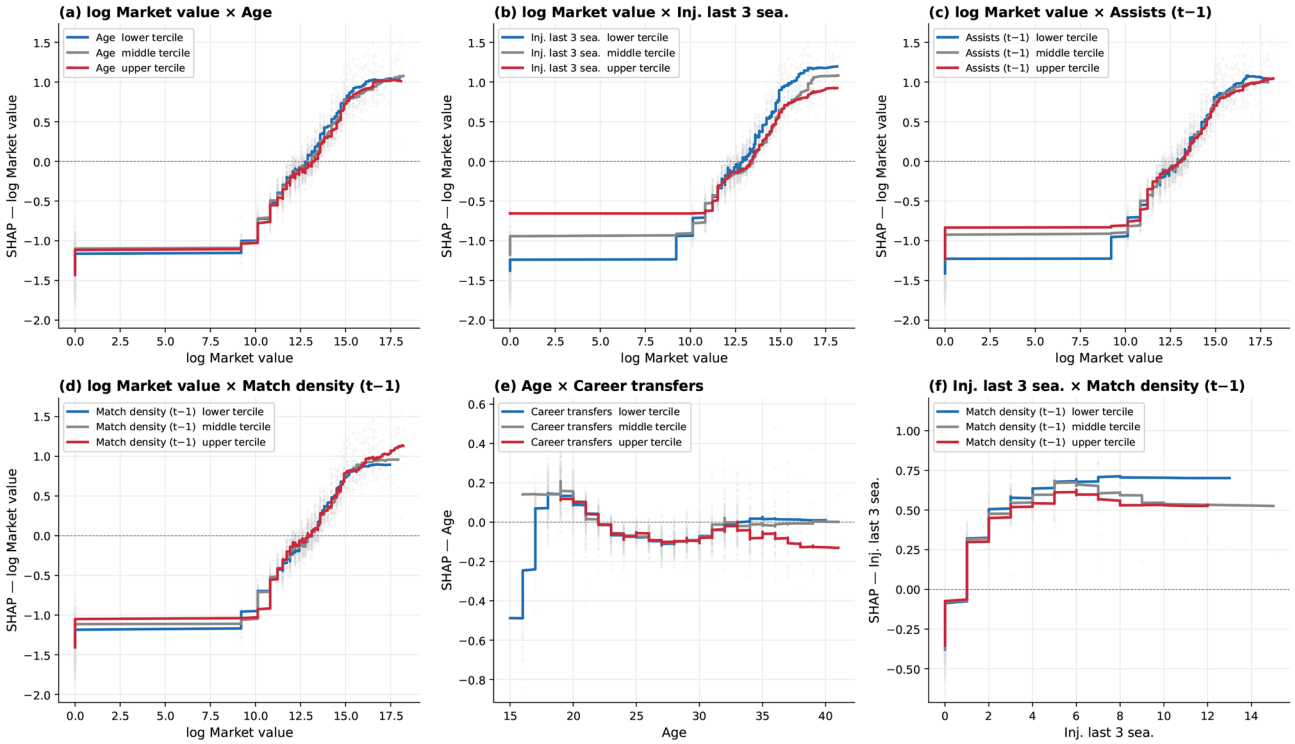



**Fig. 11**. SHAP interaction plots for six feature pairs, with the interaction variable stratified into lower (blue), middle (grey), and upper (red) terciles. (**a**) `log Market value` × `Age`; (**b**) `log Market value` × `Inj. last 3 sea.`; (**c**) `log Market value` × `Assists (t-1)`; (**d**) `log Market value` × `Match density (t-1)`; (**e**) `Age` × `Career transfers`; (**f**) `Inj. last 3 sea.` × `Match density (t-1)`.

Taken together, the interaction analysis identifies `log_market_value` as the hub of the interaction network: four of the six panels involve market value as one member of the pair, underscoring its centrality not only as a main effect but also as a moderator of other risk pathways.

*Sample-level patterns and individual decision pathways*

Figure 12 presents the SHAP heatmap for 500 test-set instances, ordered by predicted injury probability $\hat{p}$ from high-risk (left) to low-risk (right). The top curve shows $\hat{p}$ declining smoothly from approximately 0.9 to near 0. The most salient pattern is the sharp colour transition in `log Market value`: the left portion is dominated by intense red (positive SHAP), abruptly shifting to deep blue (negative SHAP) near the midpoint, confirming that market value is the primary separator between risk strata. The second and third rows—`Inj. last 3 sea.` and `All prior injuries`—display prominent red patches concentrated among the highest-risk instances (left), with increasing blue dominance toward the right. Notably, several instances in the mid-right portion of the heatmap (moderate-to-low predicted risk) exhibit localised red patches in `Prior muscle inj.` and `Cum. days missed`, identifying a subgroup whose low market visibility masks a genuine injury recurrence burden. Among prior-season exposure features, `Assists (t-1)` and `Minutes (t-1)` show mixed red-blue patterns throughout, while `Sub out (t-1)` and `Starter ratio (t-1)` display subtle blue bands among high-risk instances, consistent with the risk-attenuating association of rotation management identified in the dependence analysis.

Figure 13 displays the SHAP decision paths for 50 representative instances (25 highest-risk shown as dashed red/pink lines and 25 lowest-risk shown as solid blue lines in the test set). All paths originate from the base value and progressively diverge as each feature's contribution is sequentially incorporated from top (least important) to bottom (most important). The most consequential bifurcation occurs at `log Market value` (bottom): high-risk paths shift rightward to approximately $+3$ log-odds units, while low-risk paths shift leftward to approximately $-5$ units. Above the market value layer, `Inj. last 3 sea.` and `All prior injuries` produce the second-largest separation, with injury history features amplifying risk among already high-risk instances. Below approximately the tenth-ranked feature (from the bottom), paths exhibit minimal further separation, confirming that the upper portion of the feature hierarchy accounts for the vast majority of predictive differentiation.

Figure 14 decomposes six individual predictions representing archetypal risk profiles. Case (a) presents the highest-risk instance in the test set: a 31-year-old veteran ($p = 95.6\%$, $f(x) = 3.079$) whose risk is driven primarily by `log Market value` ($+0.91$), reflecting high visibility and competitive exposure, with substantial reinforcement from `Inj. last 3 sea.` ($+0.81$), `Muscle inj. last 3 sea.` ($+0.29$

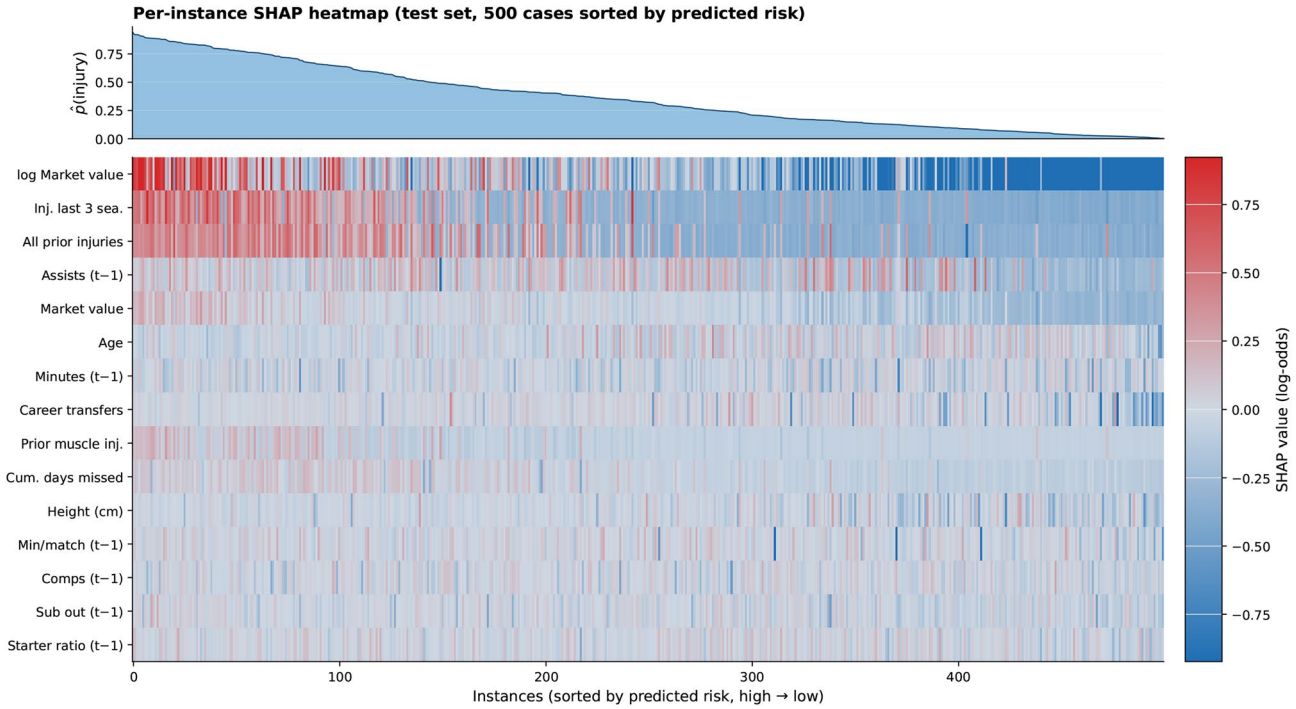



**Fig. 12**. Per-instance SHAP heatmap for 500 test-set instances ordered by predicted injury probability $\hat{p}(\text{injury})$ (top curve) from high-risk (left) to low-risk (right). The top 16 features are displayed individually. Colour encodes the signed SHAP value in log-odds units (red = positive contribution toward injury, blue = negative contribution).

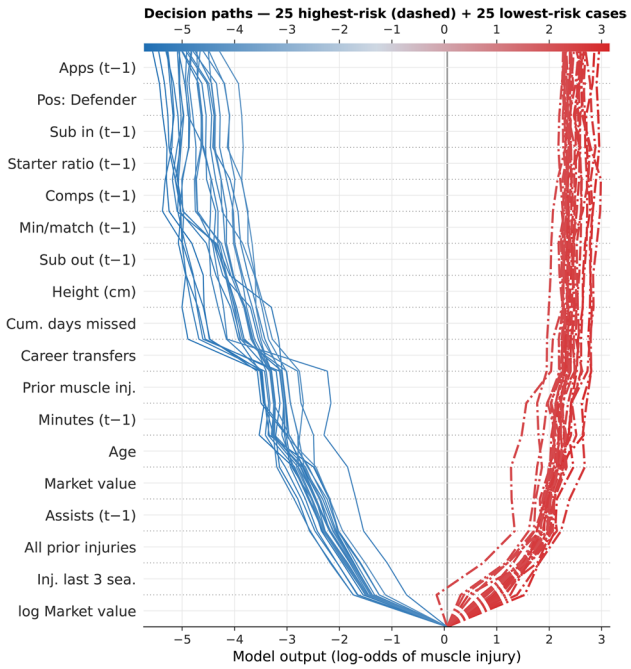



**Fig. 13**. SHAP decision paths for 25 highest-risk (dashed red) and 25 lowest-risk (solid blue) instances. All paths originate from the base value and accumulate feature contributions from top to bottom (most important features at bottom). The horizontal axis represents model output in log-odds of muscle injury. The top colour bar indicates the final predicted log-odds.

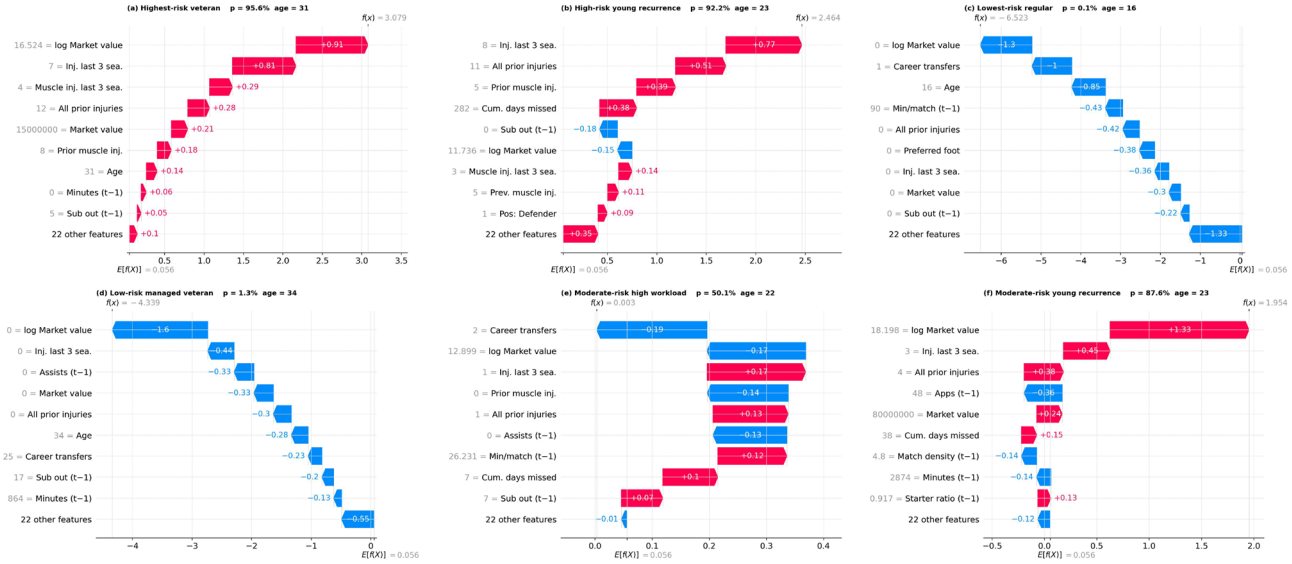



**Fig. 14**. SHAP waterfall plots for six representative cases spanning the full risk spectrum. (**a**) Highest-risk veteran: age 31, $p = 95.6\%$. (**b**) High-risk young player with injury recurrence: age 23, $p = 92.2\%$. (**c**) Lowest-risk young player: age 16, $p = 0.1\%$. (**d**) Low-risk veteran with low visibility: age 34, $p = 1.3\%$. (**e**) Decision-boundary case: age 22, $p = 50.1\%$. (f) High-visibility young player with recurrence: age 23, $p = 87.6\%$. $E[f(X)] = 0.056$ denotes the base value (population mean log-odds).

), and `All prior injuries` (+0.28). Every major feature contributes positively, illustrating a convergent high-risk profile. Case (b) illustrates a mechanistically distinct pathway: a 23-year-old player whose predicted probability reaches 92.2% despite a moderately negative market value contribution (`log Market value`: −0.15). The dominant risk drivers are injury recurrence features: `Inj. last 3 sea.` (+0.77), `All prior injuries` (+0.51), `Prior muscle inj.` (+0.39), and `Cum. days missed` (+0.38). This case demonstrates that severe injury accumulation can overwhelm the otherwise risk-attenuating association of lower market visibility.

Case (c) depicts the lowest-risk instance ($p = 0.1\%$, $f(x) = -6.523$): a 16-year-old player in whom all major features contribute negatively. `log Market value` (−1.3), `Career transfers` (−1.0), and `Age` (−0.85) dominate the risk-attenuating profile, with zero injury history further suppressing risk. This case represents a young player at the very beginning of a professional career, for whom the model assigns near-zero injury probability.

Case (d) presents a low-risk veteran ($p = 1.3\%$, age 34) whose risk remains low despite advanced age (`Age`: −0.28) primarily because of zero market visibility (`log Market value`: −1.6), zero injury history (`Inj. last 3 sea.`: −0.44), and zero prior-season assists (`Assists (t-1)`: −0.33). The `Sub out (t-1)` contribution (−0.20) suggests that substitution management further attenuates risk. This profile typifies a lower-league veteran with minimal public injury documentation.

Two additional cases illustrate intermediate risk profiles that are arguably more representative of practical decision-making contexts. Case (e) presents a 22-year-old at the exact decision boundary ($p = 50.1\%$, $f(x) \approx 0$ ), where positive and negative SHAP contributions nearly cancel. A single recent injury (`Inj. last 3 sea.`: +0.17) and modest cumulative days missed (+0.10) are offset by negative contributions from `Career transfers` (−0.19) and `log Market value` (−0.17). This balanced profile illustrates the tipping-point region where small changes in any feature could shift the prediction. Case (f) depicts a 23-year-old high-visibility player ($p = 87.6\%$) whose risk is driven primarily by `log Market value` (+1.33) alongside injury recurrence (`Inj. last 3 sea.`: +0.45; `All prior injuries`: +0.38). Interestingly, high prior-season workload features partially attenuate risk (`Apps (t-1)`: −0.36; `Match density (t-1)`: −0.14), suggesting a "healthy performer" effect in which sustained high-level participation indicates robust physical condition.

## Discussion

### Comparison with existing literature

The proposed framework achieved an AUC-ROC of 0.812 (95% CI: 0.804–0.820) under a strictly prospective temporal hold-out design, comparing favourably with prior ML-based injury prediction studies. In the most directly comparable public-data study, Diniz et al.[18] reported an AUC of 0.74 using random forests on 209 Brazilian league players—a gap that likely reflects both the substantially larger sample size and the MSFS-based feature engineering pipeline employed here. Among proprietary-data studies, Ye et al.[16] achieved AUC 0.89 using time-series image encoding with GPS-derived data; however, that study employed random rather than temporal data splitting, making direct comparison difficult, as our validation strategy comparison demonstrates that random splitting inflates AUC by approximately 3.1 percentage points. When evaluated under the same

random splitting protocol, our framework achieves AUC 0.843, which is comparable to GPS-based approaches while relying exclusively on publicly accessible data. In cross-sport comparisons, Pareek et al.[8] achieved AUC 0.84 with XGBoost on 2103 NBA players, identifying prior injury history and age as dominant predictors—consistent with the injury history and age-squared features that rank among our top-ten SHAP contributors.

The performance advantage of TabNet over XGBoost is modest (AUC difference: 0.6 percentage points under temporal hold-out). The controlled optimisation comparison demonstrates that this gap persists when both models are tuned with the same algorithm (DBA or Optuna), confirming that it reflects TabNet's attention mechanism rather than preferential hyperparameter tuning. This aligns with the benchmark literature: Grinsztajn et al.[29] demonstrated that tree-based models remain highly competitive on medium-dimensional tabular data, while McElfresh et al.[30] found that deep architectures may offer marginal advantages with large samples and complex feature interactions. We therefore recommend that practitioners prioritise the MSFS pipeline and thorough hyperparameter optimisation, while the choice between TabNet and XGBoost may be guided by computational constraints and deployment requirements. TabNet's intrinsic attention masks offer an additional interpretability channel (Spearman $\rho = 0.85$ with SHAP rankings), though SHAP provides equivalent post-hoc interpretability for any model.

## Market value as a proxy for reporting completeness

The dominance of log-transformed market value as the single most important *individual* feature in SHAP attributions warrants cautious interpretation. However, a critical finding of the prospective design is that when features are aggregated by category, Injury History emerges as the most important group (33%), surpassing Market & Career (29%). This reversal compared with same-season designs—where market-related features typically dominate—suggests that the prospective temporal design substantially reduces the confounding influence of reporting bias. While market value still encodes genuine risk-relevant information—higher-valued players typically face greater competitive intensity, more congested schedules, and elevated physical demands—the elevation of injury history to the top category confirms that the model captures genuine recurrence risk rather than primarily learning documentation patterns.

The sensitivity analysis partially disentangles these effects. When all market valuation features are excluded, AUC-ROC decreases from 0.812 to 0.786, indicating that market value contributes meaningfully but that the remaining features—particularly injury history and prior-season workload—retain substantial predictive capacity independently. The stratified analysis reveals that model performance in top-tier leagues (AUC 0.831) exceeds that in lower-tier leagues (0.772), consistent with differential reporting completeness. These findings suggest that the model's predictive capability reflects a mixture of genuine injury risk factors and reporting bias, and practitioners should interpret individual risk scores with this caveat in mind, particularly for players in leagues with less comprehensive injury documentation. The challenge of extracting genuine risk signals from unevenly documented observations is conceptually analogous to learning from incomplete supervision in other domains, where methods for inferring reliable learning signals from partial and biased annotations have shown that meaningful predictions remain achievable despite systematically missing labels[50]. Future work integrating causal inference frameworks, or developing league-specific calibration layers that account for differential reporting completeness, could further disentangle the specific mechanisms through which market value relates to injury risk.

## Practical implications

The seven-layer SHAP analysis provides actionable guidance at multiple organisational levels. At the squad management level, the identification of prior injury burden as a multiplicative risk amplifier supports evidence-based decisions regarding player acquisition, contract negotiations, and pre-season medical screening protocols—for example, a transfer target with three or more muscle injuries in the preceding three seasons would be flagged for enhanced medical evaluation before contract completion. At the match-day level, the association between prior-season substitution patterns and attenuated risk elevation provides quantitative support for rotation strategies, particularly during congested fixture periods such as mid-season tournament phases. At the individual player level, SHAP waterfall plots enable medical staff to identify the specific factors contributing to an individual player's risk score: a player whose elevated risk is driven primarily by cumulative injury burden (as in Case b of Fig. 14) would warrant a different preventive protocol from one whose risk stems predominantly from high market visibility and match exposure (as in Case f). However, these applications should be understood as decision-support tools that complement, rather than replace, clinical judgement and proprietary physiological monitoring.

The two-stage risk architecture identified through category-level SHAP analysis—in which Market/Career features establish a baseline risk stratum while Injury History and interaction features act as multiplicative amplifiers—suggests that prevention strategies should operate on both dimensions. Managing fixture load and rotation for high-exposure players addresses the first stage, whereas targeted rehabilitation protocols and return-to-play monitoring address the second.

## Limitations

Several limitations should be acknowledged. First, despite the strictly prospective feature construction design, the reliance on publicly available data means that the model operates on proxy indicators (match statistics, market valuations, publicly reported injury records) rather than direct physiological measurements such as GPS tracking data, neuromuscular screening results, or subjective wellness scores. The framework therefore models observable injury risk correlates rather than underlying biological mechanisms, and its predictive ceiling is inherently constrained by the information content of public data.

Second, injury reporting on Transfermarkt varies in completeness across leagues, tiers, and player visibility levels. The model's performance and feature importance rankings may not generalise to settings with different reporting practices. Application to proprietary club-level data, where injury ascertainment is standardised and complete, would likely alter both the model's predictive performance and the relative importance of features such as market value.

Third, while the temporal hold-out design prevents future information from informing past predictions, the model assumes stationarity in the relationship between predictors and outcomes over time. The documented temporal trend in injury incidence (rising from 3.3% in 2010/11 to 10.9% in 2024/25) may reflect genuine epidemiological changes, evolving reporting practices, or both, and model recalibration would be necessary for sustained deployment.

Fourth, the binary target variable (any muscle injury vs. none within a season) does not capture injury timing, severity, or recurrence within a season. A finer-grained outcome definition—such as time-to-first-injury or injury count—would enhance clinical utility but requires more granular temporal data than Transfermarkt currently provides.

Fifth, external validation on an independent cohort (e.g., proprietary club data with standardised injury surveillance) was not performed. While the temporal and geographic diversity of the dataset provides internal validation evidence, and the league-stratified analysis offers partial evidence of cross-context performance, true external validation remains an important direction for future work. A league-level hold-out design (e.g., training on European leagues and testing on South American leagues) would provide a stronger test of cross-geographic generalisability.

Sixth, the SHAP-based interpretability analysis identifies associations rather than causal effects. Findings such as the association between substitution patterns and attenuated risk should be interpreted as observational correlations that generate testable hypotheses for prospective intervention studies, not as evidence that specific workload management strategies causally reduce injury risk.

Seventh, the cost-sensitive learning strategy applies uniform inverse-frequency weighting to all positive examples, treating all injury cases as equally difficult to classify. In practice, positive examples in this dataset are heterogeneous in difficulty: injuries to high-visibility players in major leagues with extensive documentation and clear risk factor profiles are substantially easier to predict than injuries to lower-league players with sparse records and ambiguous feature patterns. This uniform weighting may allocate excessive model capacity to already-tractable cases while providing insufficient learning signal for the hardest ones, potentially contributing to the relatively modest AUC-PR (0.274) despite the strong AUC-ROC (0.812). Difficulty-aware training strategies—such as curriculum learning approaches that first stabilise decision boundaries on well-documented cases before progressively incorporating harder examples[51], or focal loss formulations that down-weight easily classified instances—represent a promising direction for improving precision-recall performance, particularly in the lower-visibility strata where the model is most likely to underperform.

### Future directions

Future work should pursue several extensions: integrating multi-source data by fusing public transfer-market features with proprietary training-load metrics where available, potentially leveraging cross-modal knowledge distillation techniques[48] to bridge heterogeneous data sources; refining temporal resolution from season-level to match-week-level predictions; extending the framework to other injury types and team sports where public performance data are available; implementing league-level hold-out validation to assess cross-geographic generalisability; developing causal inference approaches (e.g., instrumental variable or difference-in-differences designs) to disentangle reporting bias from genuine risk; and developing a real-time decision support tool that operationalises the SHAP-based individual risk decompositions for club medical and coaching staff.

## Conclusions

This study proposed an interpretable machine learning framework integrating TabNet with multi-method stability-weighted feature selection (MSFS) and detective behaviour algorithm (DBA) optimisation for predicting muscle injuries in professional football using exclusively public transfer-market data. A strictly prospective prediction design was implemented in which all features are constructed from prior-season data, with chronological train-test partitioning ensuring temporal integrity. Applied to 438,219 player-season observations spanning 15 seasons and 32 leagues, the MSFS pipeline achieved 46% dimensionality reduction (from 57 to 31 features) while preserving cross-validation stability (median $\phi_j = 0.90$). The DBA-optimised TabNet classifier attained an AUC-ROC of 0.812 (95% CI: 0.804–0.820) under temporal hold-out validation, outperforming nine baseline and literature-derived models under identical evaluation conditions. Direct comparison with Bayesian optimisation confirmed that DBA provides competitive hyperparameter tuning, while the controlled architecture comparison demonstrated that TabNet's advantage over XGBoost is attributable to its attention mechanism rather than preferential optimisation. A seven-layer SHAP interpretability analysis revealed that under the prospective design, Injury History emerges as the most important feature category (33%), surpassing Market & Career (29%)—a reversal from typical same-season designs that confirms the model captures genuine recurrence risk. A two-stage risk architecture was identified in which Market/Career features establish baseline risk and Injury History features act as multiplicative amplifiers among predisposed individuals. Sensitivity analyses excluding market value features quantified the contribution of reporting bias, demonstrating that the remaining features retain substantial predictive capacity independently (AUC 0.786). Individual-level waterfall decompositions indicated that prior-season substitution patterns are associated with attenuated age-related risk elevation, offering testable hypotheses for prospective intervention research. These results demonstrate that large-scale, publicly accessible transfer-market data—combined with rigorous temporal validation and careful

treatment of reporting bias—can support meaningful and interpretable injury risk stratification, broadening access to predictive tools for clubs, researchers, and governing bodies across the professional football ecosystem.

## Data availability

The data used in this study were sourced from the publicly accessible platform Transfermarkt (https://www.transfermarkt. com). The processed dataset and complete analysis pipeline (including data preprocessing, feature engineering, model training, and SHAP analysis code) will be made publicly available upon acceptance at https://github.com/[anonymised-for-review].

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## Author contributions

W.R. conceived the study, developed the methodology, wrote the software, curated the data, conducted the formal analysis and investigation, prepared the original draft, and produced the visualisations.

## Declarations

## Competing interests

The authors declare no competing interests.

## Additional information

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